4 Cell Communication and Cell Cycle MCQ

1.

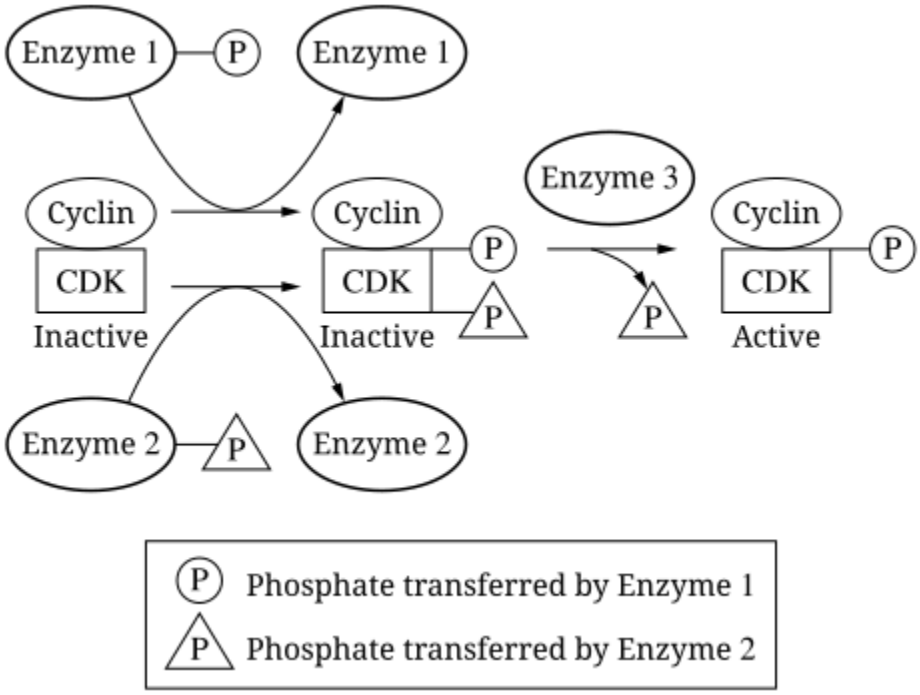


Figure 1. Activation of a cyclin-CDK complex

The regulation of a certain cyclin-CDK complex involves the activity of three enzymes (Figure 1), and the activation of this complex is required for cells to progress through the cell cycle.

Based on Figure 1, which of the following describes a cell in which cell cycle progression continues?

(A) A cell in which enzyme 1 is functional, and enzyme 2 and enzyme 3 are nonfunctional ✓

(B) A cell in which enzyme 1 is nonfunctional, and enzyme 2 and enzyme 3 are functional

(C) A cell in which enzyme 1 and enzyme 2 are functional, and enzyme 3 is nonfunctional

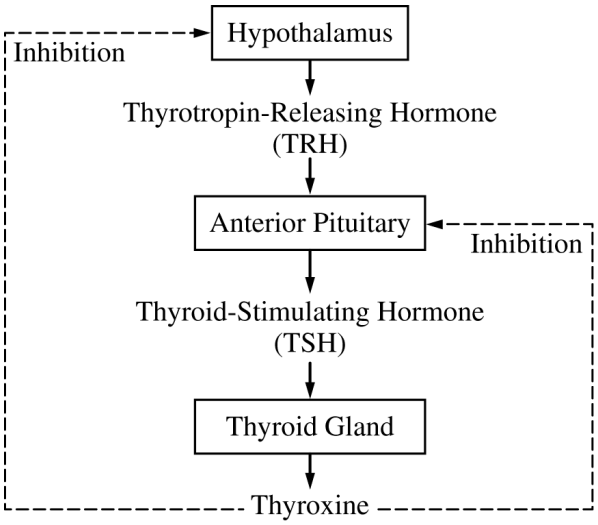
(D) A cell in which enzyme 1 and enzyme 3 are nonfunctional, and enzyme 2 is functional

Answer A

Correct. Enzyme 1 is required to transfer a phosphate group to the cyclin-CDK complex, making it active. Enzyme 2 inactivates the cyclin-CDK complex by transferring a second phosphate group to it, and enzyme 3 removes the phosphate group transferred by enzyme 2 to again activate the cyclin-CDK complex. If only enzyme 1 is functional, then the cyclin-CDK complex is active and can induce cell cycle progression.

4 Cell Communication and Cell Cycle MCQ

2.



The diagram above illustrates feedback control as exerted by the hormone thyroxine. Following surgical removal of the thyroid gland, the level of TSH in the blood will increase. Which of the following best explains this increase?

(A) Residual blood thyroxine, from prior to thyroid gland removal, will bind to cells in the anterior pituitary, signaling more TSH secretion.

(B) Thyroxine will remain bound to thyroxine receptors on various body cells, and these body cells will secrete additional hormones that stimulate the anterior pituitary to secrete TSH.

(C) Thyroxine that was stored in the anterior pituitary prior to thyroid gland removal will signal more TSH secretion.

(D) A decrease in thyroxine levels means a loss of inhibition to the hypothalamus and anterior pituitary, leading to increased TSH secretion. ✓

3. A human cell in prophase of mitosis can be distinguished from a human cell in prophase I of meiosis by the presence of

(A) kinetochores in the mitotic cell

(B) a spindle in the mitotic cell

(C) twice as many chromosomes in the meiotic cell

(D) half as many chromosomes in the meiotic cell

(E) paired homologous chromosomes in the meiotic cell ✓

4 Cell Communication and Cell Cycle MCQ

4. The relative amount of DNA in a cell at various stages of the cell cycle is shown in Figure 1 .

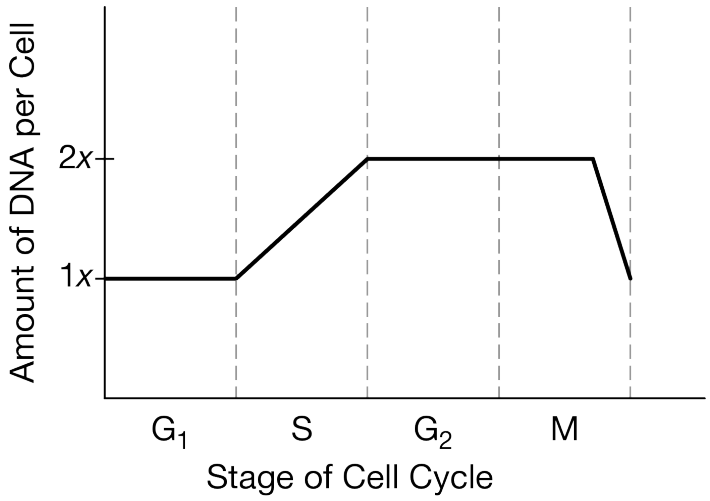


Figure 1. Amount of DNA per cell during different stages of the cell cycle, relative to the beginning of the G_1 stage

Which of the following best describes how the amount of DNA in the cell changes during M phase?

(A) The amount of DNA doubles as the DNA is replicated.

(B) The amount of DNA slightly increases as a result of new organelle synthesis.

(C) The amount of DNA does not change while the cell grows.

(D) The amount of DNA is halved as the cell divides into two daughter cells. ✓

Answer D

Correct. During mitosis, sister chromatids are separated, two nuclei form, and cytokinesis results in the formation of two new cells. Each daughter cell has the same number of chromosomes as the original parent cell.

5. Antidiuretic hormone (ADH) is important in maintaining homeostasis in mammals. ADH is released from the hypothalamus in response to high tissue osmolarity. In response to ADH, the collecting duct and distal tubule in the kidney become more permeable to water, which increases water reabsorption into the capillaries. The amount of hormone released is controlled by a negative feedback loop.

Based on the model presented, which of the following statements expresses the proper relationship between osmolarity, ADH release, and urine production?

4 Cell Communication and Cell Cycle MCQ

(A) As tissue osmolarity rises, more ADH is released, causing less water to be excreted as urine. ✓

(B) As tissue osmolarity rises, less ADH is released, causing less water to be excreted as urine.

(C) As tissue osmolarity rises, more ADH is released, causing more water to be excreted as urine.

(D) As tissue osmolarity rises, less ADH is released, causing more water to be excreted as urine.

6. Researchers have discovered details about apoptosis (programmed cell death) by studying embryologic development of a nematode worm, *Caenorhabditis elegans*. Apoptosis is a normal developmental process in *C. elegans*. They found several genes involved in apoptosis, including $ced-9$ and $ced-3$. The $ced-3$ gene was found to promote cell death, and $ced-9$ to inhibit it. The $ced-9$ gene serves as a regulator that prevents apoptosis in the absence of a signal promoting apoptosis.

Which of the following statements best justifies the claim that changes in the expression of $ced-9$ in *C. elegans* can affect regulation of apoptosis in the cell?

(A) An experiment showed that a mutation in the $ced-9$ gene led to excessive cell death in *C. elegans*. ✓

(B) An experiment showed that the $ced-9$ gene normally produces a protein that promotes excessive cell death in *C. elegans.*

(C) A mutation in $ced-3$ will cause $ced-9$ to be incorrectly transcribed.

(D) Apoptosis is dependent on a signal from the $ced-9$ gene in *C. elegans.*

Answer A

Correct. If $ced-9$ acts as a regulator and normally inhibits apoptosis, then if it loses function, more cell death will occur.

7. A researcher examining a root tip observes a plant cell with condensed sister chromatids, kinetochores with attached microtubules, and individual chromosomes that are aligned at the equatorial plate of the cell. Which of the following best explains what the next process will be in the cell?

(A) Homologous chromosomes (each with two sister chromatids) will move toward opposite poles of the cell.

(B) Paired chromatids will separate, and the new daughter chromosomes will move toward opposite poles of the cell. ✓

(C) The nuclear envelope will break down, and the spindle will begin to form.

(D) The chromatin will decondense, and the daughter cell will enter interphase.

4 Cell Communication and Cell Cycle MCQ

8. Phosphofructokinase (PFK) is a key enzyme in glycolysis. ATP is one of the two substrates for the reaction catalyzed by PFK. ATP is also an allosteric regulator of PFK. Figure 1 shows the enzyme-substrate interactions of PFK.

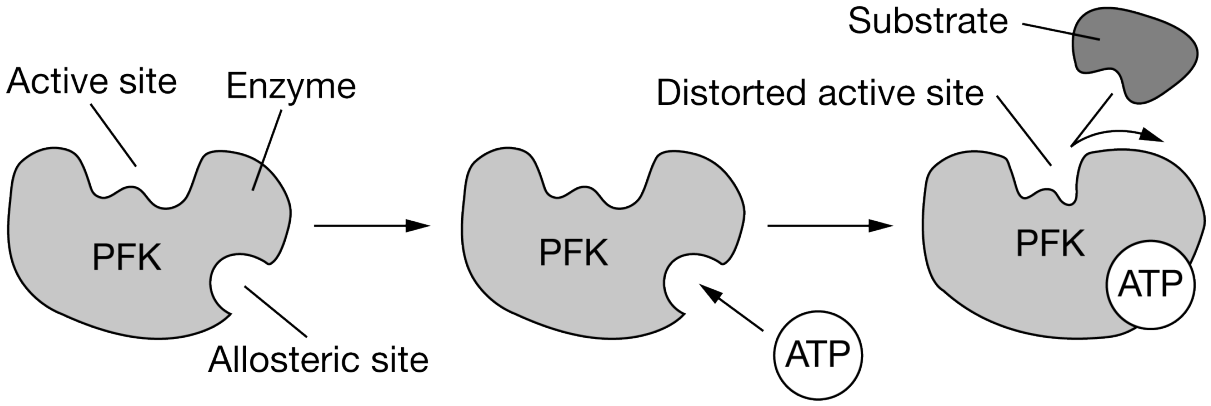


Figure 1. The enzyme-substrate interactions of PFK

A researcher found a mutation that resulted in the PFK enzyme being unable to bind ATP to the allosteric site. Which of the following best predicts the effect of the mutation?

(A) The activity of the enzyme will not be affected because the active site is not involved in substrate binding at the allosteric site.

(B) Negative feedback regulation does not occur, so the enzyme will be active when glycolysis is not needed. ✓

(C) Positive feedback does not occur, and the activity of the enzyme will decrease when glycolysis is needed.

(D) The activity of the enzyme will fluctuate independent of the ATP concentration.

Answer B

Correct. The reaction is normally inhibited when ATP binds to the allosteric site and changes the shape of the active site to the point where it can no longer bind to the substrate. The mutation will not allow this inhibition to occur, and the enzyme will be active even at high ATP concentrations.

4 Cell Communication and Cell Cycle MCQ

In response to elevated blood glucose levels, beta (β) cells in the pancreas release insulin, a regulatory hormone. Insulin signals body cells to take up glucose from the blood, which returns blood glucose levels back to normal. Type 1 diabetes is an autoimmune disorder that destroys β-cells, resulting in elevated blood glucose levels.

Researchers have proposed that diabetes could be treated by implanting human embryonic stem cells (hESCs) that have been induced to develop into β-cells (hESC-β). To test the proposed treatment, the researchers set up two groups of genetically identical mice and implanted the mice from one group with hESC-β cells. Several weeks after the hESC-β implant, both groups of mice were given a drug (STZ) that selectively destroys the naturally occurring mouse β-cells but does not affect the implanted hESC-β cells. Figure 1 shows a comparison of average blood glucose levels in both groups of mice.

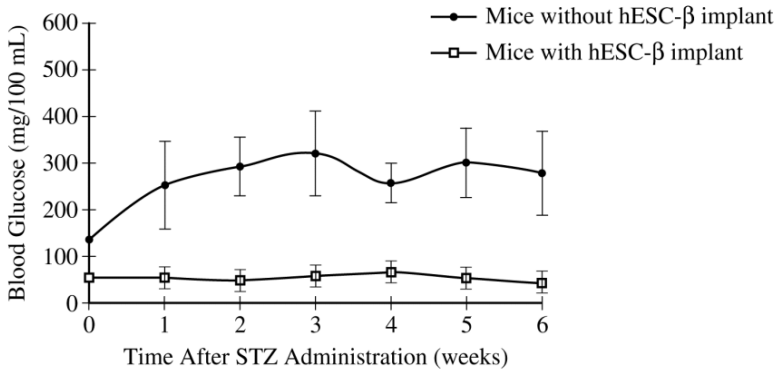


Figure 1. Average blood glucose levels in mice after STZ treatment. Error bars indicate standard deviation.

In a continuation of the experiment, the researchers removed the hESC-β implant from one of the mice 16 weeks after STZ treatment. Figure 2 shows the blood glucose levels in the mouse over the duration of the experiment.

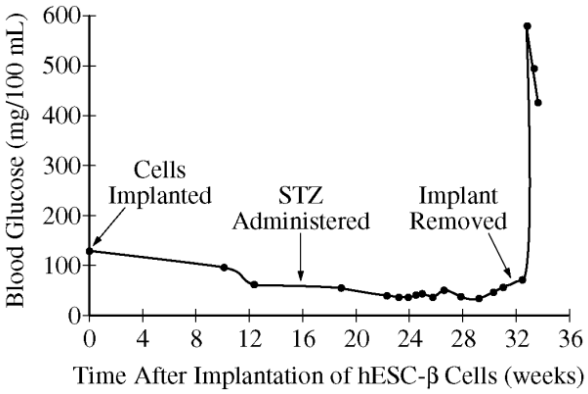


Figure 2. Blood glucose levels for an individual mouse over the duration of the experiment.

4 Cell Communication and Cell Cycle MCQ

9. Based on the data, which of the following best explains how the mice with the implanted hESC-β cells use negative feedback to return blood glucose levels to normal if blood glucose levels increase?

(A) Decreased insulin secretion by body cells → increased glucose uptake by mouse β-cells → increased insulin secretion by body cells

(B) Increased insulin production by mouse β-cells → increased glucose uptake by hESC-β cells → decreased glucose metabolism by body cells

(C) Increased insulin secretion by hESC-β cells → increased glucose uptake by body cells → decreased insulin secretion by hESC-β cells ✓

(D) Increased metabolism of glucose by hESC-β cells → differentiation of hESC-β cell into body cells → increased glucose production by body cells

10. Which of the following modifications to the experimental design would improve the reliability of the results presented in Figure 1 ?

(A) Measuring blood glucose levels in human patients who have type 1 diabetes

(B) Measuring blood glucose levels in a group of mice that were not given STZ ✓

(C) Using a drug that destroys both mouse β-cells and hESC-β cells

(D) Injecting the mice in the experiment with sucrose or fructose

11. Which of the following conclusions is best supported by the data presented in Figure 1 ?

(A) Mouse β-cells are more effective than implanted hESC-β cells at regulating blood glucose levels.

(B) Implanted hESC-β cells are effective at regulating blood glucose levels in the absence of mouse β-cells. ✓

(C) Mouse b-cells do not regulate blood glucose levels in the presence of implanted hESC-β cells.

(D) Implanted hESC-β cells cause type 1 diabetes in laboratory mice.

12. In contrast to type 1 diabetes, where there is no insulin production, type 2 diabetes is characterized by a failure of body cells to respond normally to insulin. Based on the results of the experiment, which of the following best predicts the effectiveness of using hESC-β implantation to treat type 2 diabetes?

(A) The treatment would fail because insulin produced by implanted hESC-β cells will not stimulate glucose uptake by body cells. ✓

(B) The treatment would fail because the implanted hESC-β cells would require more glucose than the body can produce.

(C) The treatment would succeed because the implanted hESC-β cells can express the gene that codes for human insulin.

(D) The treatment would succeed because the body cells would secrete more insulin in the presence of the implanted hESC-β cells.

13. Based on the data in Figure 2, which of the following best justifies the researchers' claim that hESC-β implants can functionally replace the mouse's naturally occurring β-cells?

4 Cell Communication and Cell Cycle MCQ

(A) The mouse with an hESC-β implant demonstrated a decrease in blood glucose levels for the duration of the experiment.

(B) The mouse with an hESC-β implant demonstrated an increase in blood glucose levels before STZ treatment.

(C) The mouse with an hESC-β implant successfully regulated blood glucose levels by regenerating its naturally occurring β-cells.

(D) The mouse with an hESC-β implant successfully regulated blood glucose levels after STZ treatment but not after the implant was removed. ✓

4 Cell Communication and Cell Cycle MCQ

Insulin, a hormone secreted by pancreatic cells, stimulates glucose uptake in skeletal muscle cells by mobilizing glucose transporter proteins (GLUT4) to the plasma membrane. As depicted in Figure 1, binding of insulin to the insulin receptor triggers an intracellular signaling cascade in which certain molecules activate other molecules in a relay of the hormone signal to cell targets. One outcome of the signaling cascade is mobilization of GLUT4 from vesicle storage sites in the cytoplasm to sites at the cell surface, where GLUT4 allows glucose to enter the cell.

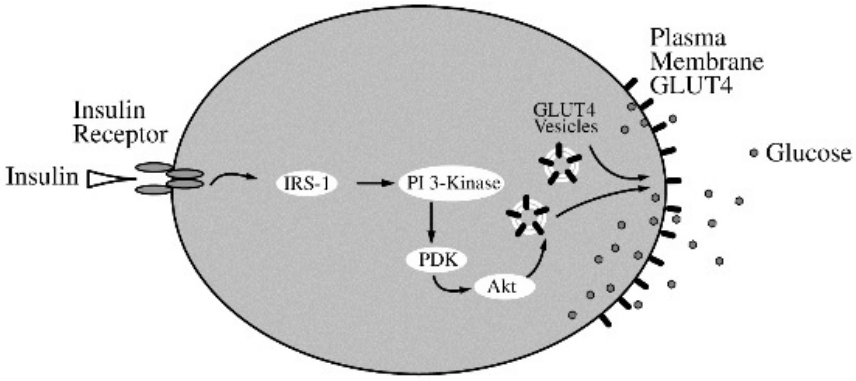


Figure 1. Insulin signaling in muscle cells

In type 2 diabetes, the cellular response to insulin is disrupted, and individuals with type 2 diabetes cannot properly regulate their blood glucose levels. In an investigation of the insulin signaling pathway, samples of skeletal muscle were isolated from individuals who have type 2 diabetes and from individuals who do not. The results of several experiments that were performed on the muscle samples are shown in Figure 2, Figure 3, and Figure 4.

4 Cell Communication and Cell Cycle MCQ

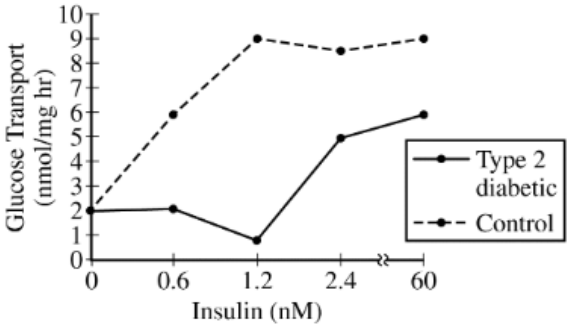


Figure 2. Insulin-stimulated glucose uptake

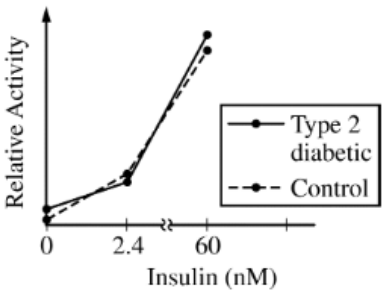


Figure 3. Insulin receptor activation

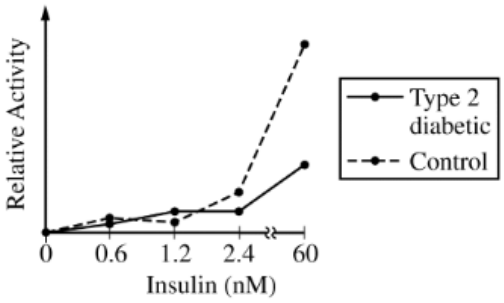


Figure 4. IRS-1 activation

14. Based on the experimental results, which of the following explains the most likely defect in muscle cells of patients with type 2 diabetes?

(A) Insulin receptor proteins do not reach the cell surface.

(B) Insulin does not activate its receptor.

(C) IRS-1 activation is reduced at high insulin concentrations. ✓

(D) GLUT4 blocks glucose from entering cells.

15. Which of the following is a valid interpretation of the experimental results that explains how individuals with type 2 diabetes differ from individuals without diabetes?

(A) The relatively low levels of glucose uptake in individuals with type 2 diabetes indicate that mobilization of GLUT4 to the cell surface is reduced in muscle cells of those individuals. ✓

(B) The relatively low levels of glucose uptake in individuals with type 2 diabetes indicate that no functional GLUT4 protein is produced in the muscle cells of those individuals.

(C) The absence of activated insulin receptors in individuals with type 2 diabetes indicates that no insulin is secreted by the pancreatic cells of those individuals.

(D) The absence of activated IRS-1 in individuals with type 2 diabetes indicates that no functional insulin receptor protein is produced in the muscle cells of those individuals.

4 Cell Communication and Cell Cycle MCQ

16. Based on the information presented, which of the following genetic changes in an individual without diabetes is most likely to result in a disrupted cellular response to insulin signaling similar to that of an individual with type 2 diabetes?

(A) A deletion in the gene encoding the insulin receptor that removes only the cytoplasmic domain of the protein ✓

(B) Duplication of the gene encoding a PI-3 kinase that results in synthesis of a muscle-specific variant of the enzyme in skin cells as well as in muscle cells

(C) A mutation in the gene encoding IRS-1 that causes the protein to be active in muscle cells even in the absence of insulin signaling

(D) Insertion of a small segment of DNA into the promoter of the *Glut4* gene that results in increased synthesis of GLUT4 proteins in muscle cells

4 Cell Communication and Cell Cycle MCQ

Excess intracellular iron is toxic to cells (iron-induced toxicity). Ferritin is an intracellular iron storage protein that binds excess iron. The presence of ferritin can protect cells from iron-induced toxicity.

In an experiment to investigate the effects of dietary iron intake on ferritin synthesis, rats were given food containing different amounts of iron. Subsequently, the levels of ferritin protein in the liver were measured. The results are shown in Figure 1.

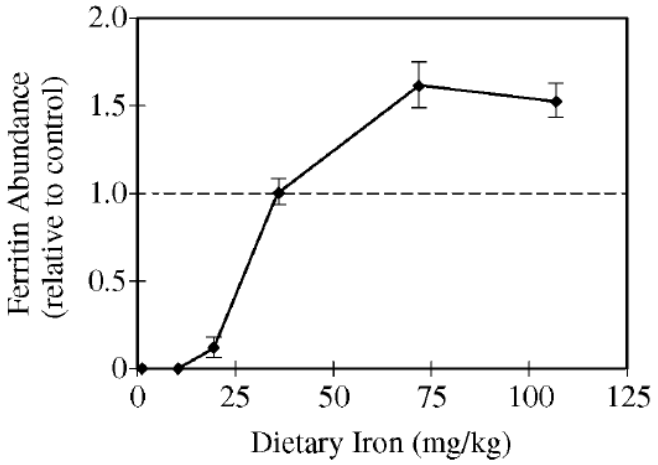
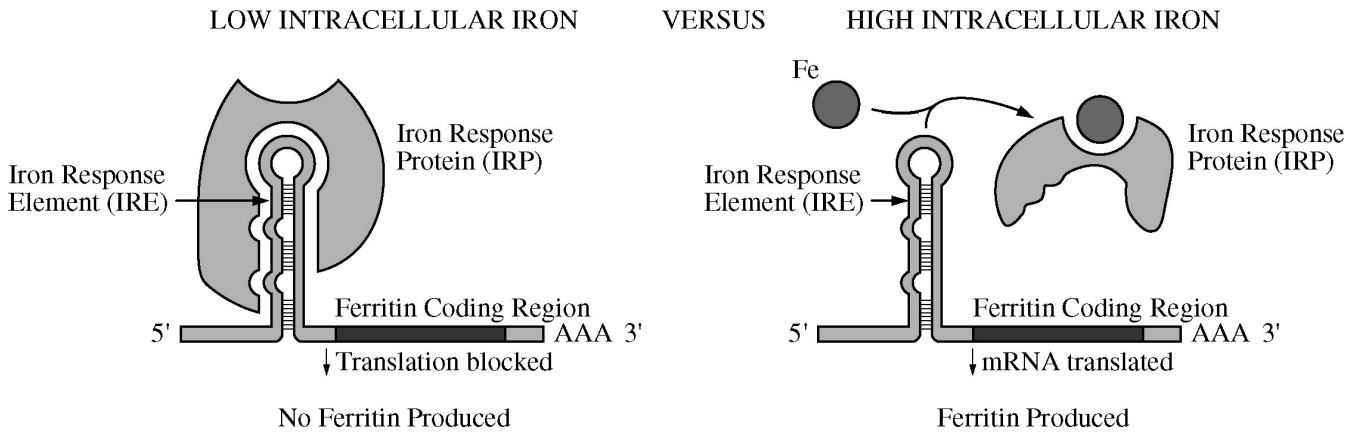


Figure 1. Effects of dietary iron on ferritin levels in rat liver

Based on these and other data, researchers have developed the following model demonstrating how ferritin synthesis is regulated by iron. When iron levels are low, a repressor of translation, iron response protein (IRP), binds to an iron response element (IRE), which is a stem-loop structure near the 5¢ end of ferritin mRNA. When iron levels are high, intracellular iron binds to the IRP, and the iron-IRP complex dissociates from the IRE, permitting ribosomes to proceed with the translation of ferritin mRNA. Figure 2 represents the model of the regulation of ferritin mRNA translation by iron.



4 Cell Communication and Cell Cycle MCQ

Figure 2. Model of regulation of ferritin synthesis by iron

17. Based on the model of ferritin synthesis presented in Figure 2, which of the following describes the role of feedback on the control of intracellular iron levels?

(A) A decrease in iron levels activates the IRP. The IRP in turn activates iron transport proteins in the cell membrane, thereby returning free iron levels to normal.

(B) A decrease in iron levels activates synthesis of ferritin protein. Ferritin protein in turn releases bound iron, thereby returning free iron levels to normal.

(C) An increase in iron levels activates the IRP. The IRP in turn binds iron, thereby decreasing both free iron levels and ferritin synthesis.

(D) An increase in iron levels activates synthesis of ferritin protein. Ferritin protein in turn binds iron, thereby decreasing both free iron levels and ferritin synthesis. ✓

18. Based on the model of ferritin synthesis presented in Figure 2, which of the following best describes the mechanism whereby iron most likely regulates ferritin production?

(A) Translation occurs under low intracellular iron concentration when the IRP recruits ribosomes to the ferritin mRNA.

(B) Translation occurs under low intracellular iron concentration when the IRP stabilizes the stem-loop structure in the ferritin mRNA.

(C) Translation occurs under high intracellular iron concentration when the IRP-iron complex dissociates from ferritin mRNA, permitting ribosomes access to the ferritin coding region. ✓

(D) Translation occurs under high intracellular iron concentration when the IRP-iron complex brings the 5¢ end of the mRNA closer to the ferritin coding region.

4 Cell Communication and Cell Cycle MCQ

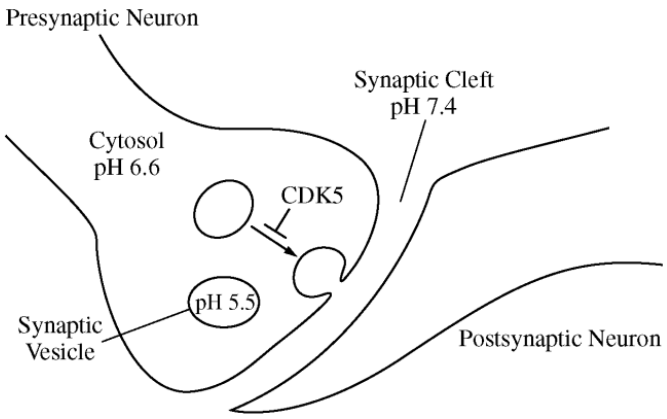


Figure 1. Model of synapse

Researchers investigating the regulation of neurotransmitter release from presynaptic neurons proposed a model (Figure 1) in which CDK5, a protein expressed in axon terminals, inhibits the movement of synaptic vesicles to the presynaptic membrane.

To test their model, the researchers used a modified version of green fluorescent protein (GFP*). In slightly alkaline conditions, GFP* exhibits a bright green fluorescence. In acidic conditions, GFP* exhibits no fluorescence. Using standard techniques, the gene encoding GFP* is easily introduced into living cells. By engineering the expression of GFP* in laboratory-cultured nerve cells, the researchers found that a bright green fluorescence was exhibited only when a presynaptic neuron was given a certain stimulus.

19. Based on the model, which of the following best explains how regulation of neurotransmitter release might increase the range of responses to a stimulus in the nervous system?

(A) In the absence of any stimulus, neurons can still release neurotransmitters.

(B) Different neurons in the same neural network can release different amounts of neurotransmitter. ✓

(C) In the depolarization phase of an action potential, postsynaptic neurons can adjust the amount of neurotransmitter bound to receptors on their surface.

(D) In the recovery phase following a stimulus, enzymes can be mobilized to degrade molecules present in the synaptic vesicles.

20. Previous experiments indicate that CDK5 is active only when attached to a protein called p35. Which of the following best predicts how p35 might play a role in regulating neuron function?

(A) Elevated intracellular levels of p35 result in increased synaptic activity.

(B) Degradation of p35 results in increased synaptic activity. ✓

(C) Reabsorption of p35 from the synaptic cleft results in increased synaptic activity.

(D) Attachment of p35 to synaptic vesicles results in increased synaptic activity.

21. Which of the following observations best supports the hypothesis that CDK5negatively regulates neurotransmitter release?

4 Cell Communication and Cell Cycle MCQ

(A) Introduction of CDK5 protein into neurons results in the movement of synaptic vesicles to the plasma membrane in the absence of any stimulus.

(B) Uptake of a gene encoding CDK5 by neurons results in the movement of synaptic vesicles to the plasma membrane in the absence of any stimulus.

(C) Suppression of CDK5 expression in neurons inhibits the movement of synaptic vesicles to the plasma membrane in response to a specific stimulus.

(D) Inhibition of CDK5 activity in neurons increases the movement of synaptic vesicles to the plasma membrane in response to a specific stimulus. ✓

22. The beta-2 adrenergic receptor is a membrane-bound protein that regulates several cellular processes, including the synthesis and breakdown of glycogen. The receptor binds specifically to the hormone epinephrine. The binding of epinephrine to the beta-2 adrenergic receptor triggers a signal transduction cascade that controls glycogen synthesis and breakdown in the cell. A simplified model of the signal transduction cascade is represented in Figure 1.

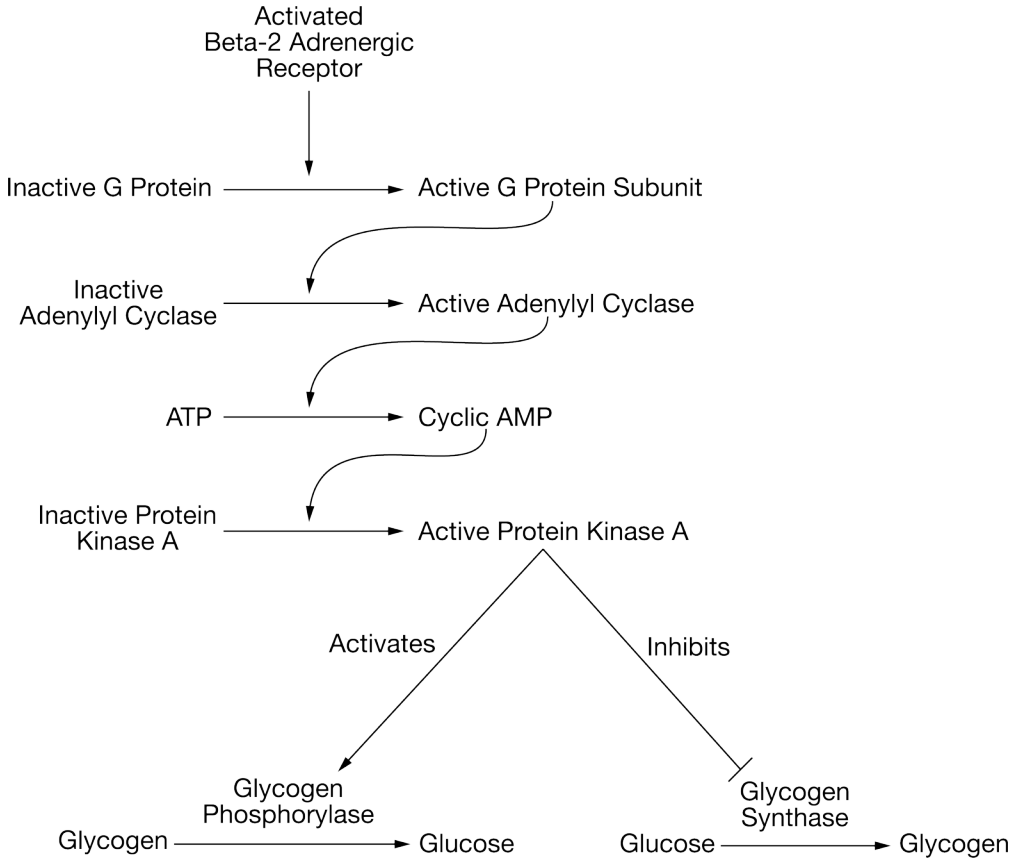


Figure 1. A simplified model of the signal transduction cascade triggered by epinephrine binding to the beta-2 adrenergic receptor

Which of the following outcomes will most likely result from the inactivation of the beta-2 adrenergic receptor?

4 Cell Communication and Cell Cycle MCQ

(A) The cellular concentration of cyclic AMP will increase.
(B) The enzymatic activity of protein kinase A will increase.
(C) The activation of glycogen phosphorylase will increase.
(D) The rate of glycogen synthesis in the cell will increase. ✓

Answer D

Correct. Based on Figure 1, activation of the beta-2 adrenergic receptor results in the stimulation of glycogen breakdown and the inhibition of glycogen synthesis. The immediate effect of inactivating the beta-2 adrenergic receptor will be an increase in the rate of glycogen synthesis in the cell because glycogen synthase will no longer be inhibited. Also, glycogen breakdown in the cell will no longer be stimulated because glycogen phosphorylase will most likely be deactivated.

4 Cell Communication and Cell Cycle MCQ

23. Blood clots are formed by a positive feedback loop. Two pathways exist, the extrinsic and intrinsic pathways, which converge during clot formation. There are many clotting factors involved, most of which are proteins. Vitamin K is required for the formation of the active form of several of the clotting factors, including Factor X. A simplified model of the blood clotting process is shown in Figure 1.

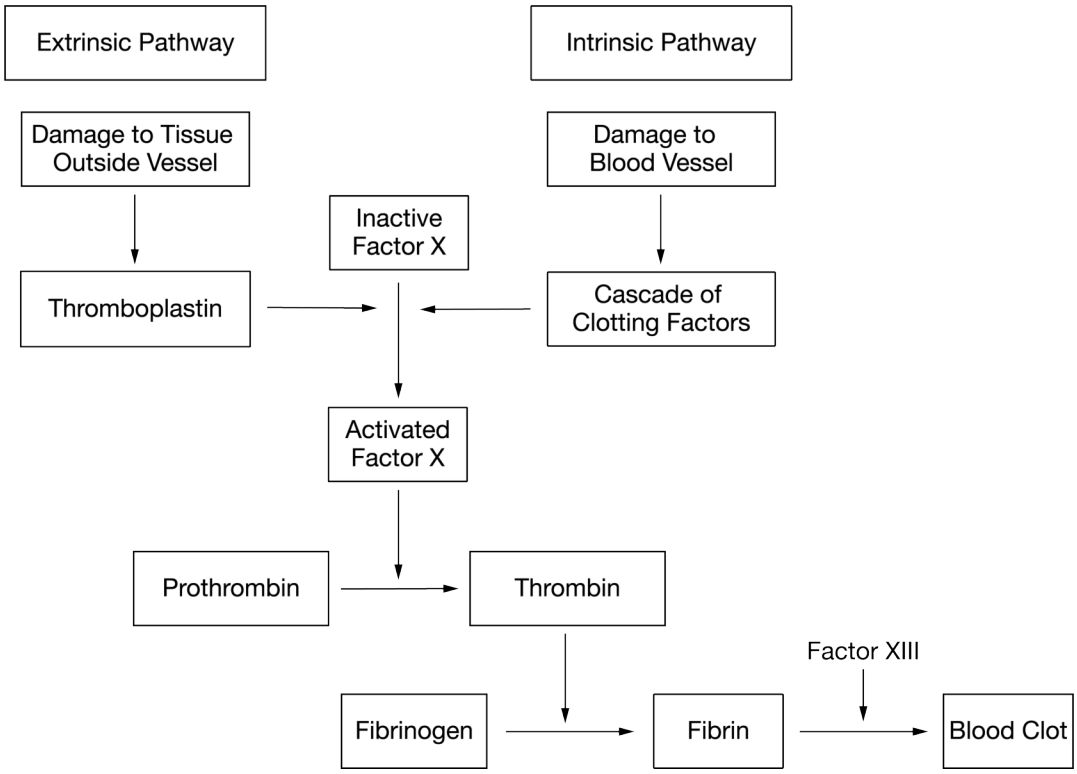


Figure 1. Simplified model of clotting cascade

Warfarin is a drug used to treat certain blood clots. Warfarin blocks the formation of the active form of vitamin K -dependent clotting factors. Based on the model, which of the following best predicts the effects of warfarin on a patient?

(A) Fibrinogen will form fibrin, but the clot will not form because Factor XIII will not be synthesized.

(B) The intrinsic pathway will take over because the clotting factors are part of that pathway.

(C) Thrombin will be converted to prothrombin because Factor X will reverse the reaction.

(D) Factor X will not be activated, which will prevent thrombin from forming. ✓

Answer D

Correct. Vitamin K is required for the synthesis of Factor X and warfarin inhibits that process; thus, clots are prevented from forming.

4 Cell Communication and Cell Cycle MCQ

24. A person's blood glucose level fluctuates during the day, as represented in Figure 1. Two hormones, insulin and glucagon, are directly involved in regulating the blood glucose level to maintain a healthy level. Insulin acts to lower the blood glucose level, and glucagon acts to increase the blood glucose level.



Figure 1. Blood glucose fluctuations of an individual

Which of the following best predicts what will happen to the blood glucose level if the person has another meal at 5 P.M.?

(A) Immediately after the meal, the blood glucose level will decrease because of the increase in glucagon levels.

(B) Immediately after the meal, the blood glucose level will increase, and then insulin will be secreted to counter the increase. ✓

(C) Several hours after the meal, the blood glucose level will increase sharply because of an increase in the amount of glucagon secreted.

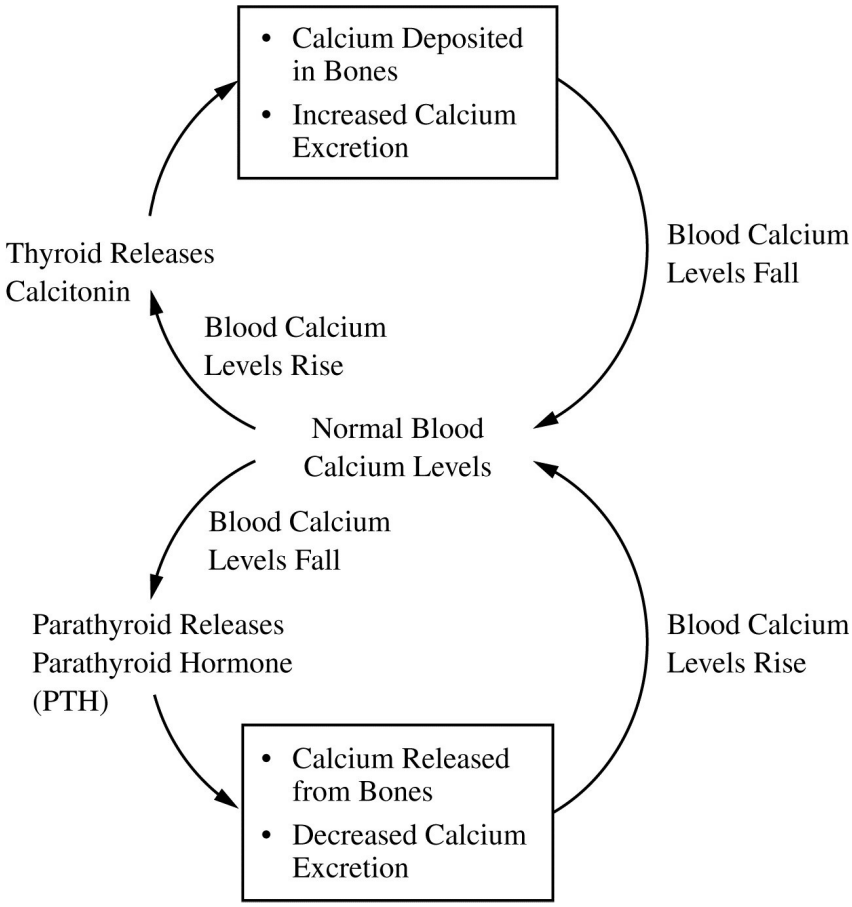
(D) The blood glucose level will not change after the 5 P.M. meal because the person has already consumed two meals and the blood glucose level has been adjusted to a steady-state level.

Answer B

Correct. The blood glucose levels increase due to the nutrients taken in. This increase is a signal for the pancreas to release more insulin that will stimulate cells to take in the excess glucose. As a result, the blood glucose level will then decrease.

4 Cell Communication and Cell Cycle MCQ

25.



The model shown in the figure represents the role of two hormones, calcitonin and parathyroid hormone (PTH), in maintaining normal blood calcium levels in humans. If a dietary change results in an increase in blood calcium concentration above normal levels, which of the following is the most likely effect on calcium homeostasis?

(A) Calcitonin levels will decline, thus stimulating the release of PTH.

(B) Calcitonin levels will rise, thus promoting the deposit of calcium into bones. ✓

(C) PTH levels will decline, thus stimulating the loss of calcium from bones.

(D) PTH levels will increase, thus preventing the release of calcitonin.

26. Cancer can result from a variety of different mutational events. Which of the following is LEAST likely to result in the initiation of a cancerous tumor?

(A) A receptor mutation results in activation of a cell-division pathway in the absence of the appropriate ligand.

(B) A mutation results in the loss of the ability to produce a tumor-suppressor protein.

(C) A defect in a cell-cycle checkpoint prevents a cell from entering the S phase. ✓

(D) At the anaphase checkpoint, separation of chromatids occurs without all centromeres being attached to kinetochore microtubules from both poles.

4 Cell Communication and Cell Cycle MCQ

27. Most cells that have become transformed into cancer cells have which of the following characteristics when compared to normal, healthy cells?

(A) Shorter cell cycle ✓

(B) More carefully regulated rates of cell division

(C) Lower rates of mitosis

(D) Higher rates of protein translation

(E) Identical DNA

28. Cyclin-dependent kinase 7 (CDK 7) binds to cyclin H and a second accessory protein, MAT 1, to form a complex of three proteins. Researchers investigating the role of CDK 7 claim that formation of this complex promotes cell cycle progression.

Which of the following supports the researchers' claim?

(A) Decreased expression of CDK 7 initiates DNA replication.

(B) Decreased expression of CDK 7 results in the formation of spindle fibers.

(C) Increased expression of CDK 7 causes cells to enter G_0.

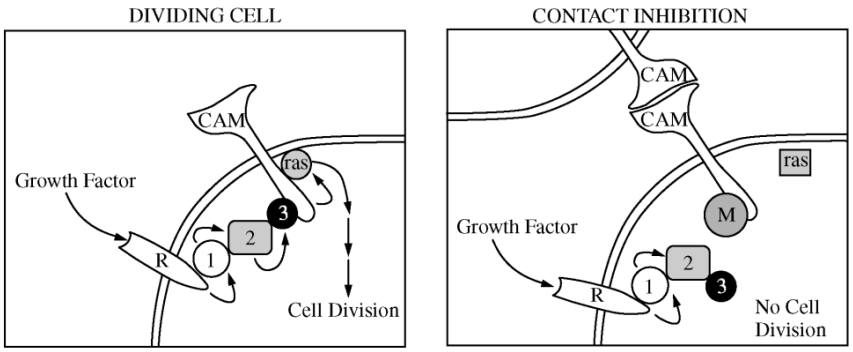
(D) Increased expression of CDK 7 is associated with certain types of cancer. ✓

Answer D

Correct. Cancer is associated with increased cell division. Increased expression of CDK 7 in certain cancers would indicate increased cell cycle progression and would therefore support the researchers' claim.

4 Cell Communication and Cell Cycle MCQ

29. Many human cells can be stimulated to divide by hormonelike growth factors that bind to receptor proteins (R) on the plasma membrane and trigger an internal signal-transduction cascade. In many cases, however, the process of contact inhibition prevents mitosis when cells are in direct contact with one another. Contact inhibition occurs when proteins called cell adhesion molecules (CAMs) interact, causing them to change shape so that the growth-factor signaling proteins that normally associate with CAMs are replaced by another protein, called M. Both pathways are depicted in the figures below.



Which of the following statements accurately uses the information presented to support the hypothesis that interruption of M function in a single body cell can result in cancer?

(A) Protein 3 will be prevented from interacting with CAMs, causing the cell cycle to stop permanently.

(B) The ras protein will remain bound to DNA, blocking expression of genes required for mitosis.

(C) Growth-factor signaling can trigger mitosis in cells that are in direct contact with other cells. ✓

(D) The receptor proteins of body cells will no longer bind to growth-factor proteins.

30. Which of the following best describes a cell that is in the G_1 stage of the cell cycle?

(A) The cell is growing in size and increasing its number of proteins and organelles. ✓

(B) The chromosomes within the cell are checked to make certain that they have been replicated correctly.

(C) The chromosomes are replicating, and the amount of DNA in the cell is doubling.

(D) The cell is in a resting stage and no longer dividing.

Answer A

Correct. The cellular activities described are activities that occur during the G_1 stage of the cell cycle.

4 Cell Communication and Cell Cycle MCQ

31. A cell culture commonly used in research was selected to study the effect of a specific virus on the timing of cell cycle phases. Two separate cultures were started, one untreated and one inoculated with the virus. Both cultures were incubated under identical conditions. After a period of time, 200 cells from each culture were observed and classified as shown in Table 1.

Table 1. Number of normal and infected cells found in three phases of the cell cycle

Phase of Cell Cycle	Untreated Cells ($n = 200$)	Virus-Infected Cells ($n = 200$)
G_0	196	10
Interphase	2	40
Mitosis	2	150

Which of the following most accurately describes an observation and an effect of the viral infection indicated by the data in Table 1?

(A) Normal cells spend 98 percent of their time cycling in and out of interphase. The virus reduces this to 5 percent of the time.

(B) Twenty percent of the virus-infected cells are in interphase. These cells are no longer part of the cell cycle.

(C) Forty percent of the virus-infected cells are in interphase. These cells are preparing for replication of genetic material.

(D) Seventy-five percent of the virus-infected cells are found in mitosis. The virus stimulates frequent cell division. ✓

Answer D

Correct. One hundred fifty out of 200 is 75 percent. The virus activates cells currently in the G_0 phase, leading to rapid cell growth and division.

4 Cell Communication and Cell Cycle MCQ

32. Figure 1 represents the relative time and sequence of the phases of the cell cycle.

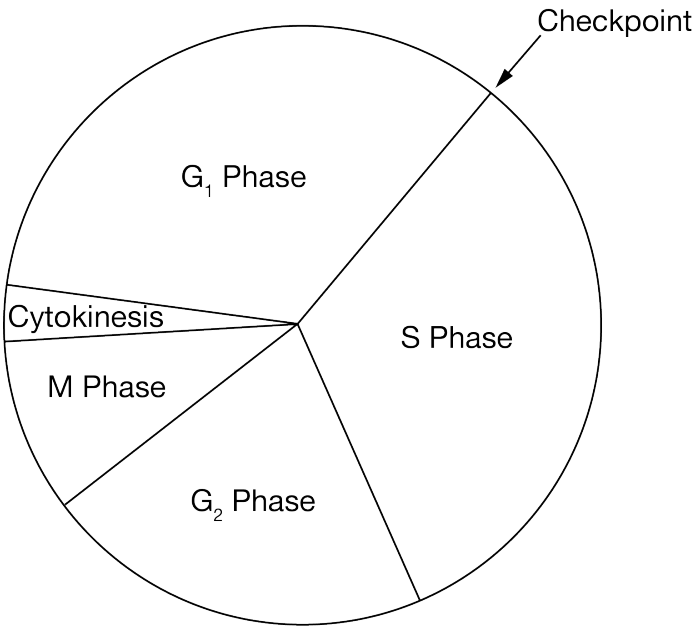


Figure 1. Representation of the cell cycle and identification of the G_1/S checkpoint

Which statement best predicts why a cell's progression through the cell cycle might be halted at the G_1/S checkpoint?

(A) Spindle fibers have not correctly attached to chromosomes.

(B) There are not enough nucleotides available to construct new DNA. ✓

(C) Damage occurred to DNA when it was being copied in G_1.

(D) Proteins necessary for M phase of the cell cycle have not been produced.

Answer B

Correct. DNA replication, which occurs during S phase, requires free nucleotides. If nucleotides are unavailable, the G_1/S checkpoint halts the cell cycle to conserve energy.

4 Cell Communication and Cell Cycle MCQ

33. Researchers performed an experiment to determine the effect of certain genetic mutations on mitosis in tropical fruit fly embryos. They determined the percentage of cells in each of four phases of mitosis as shown in Figure 1.

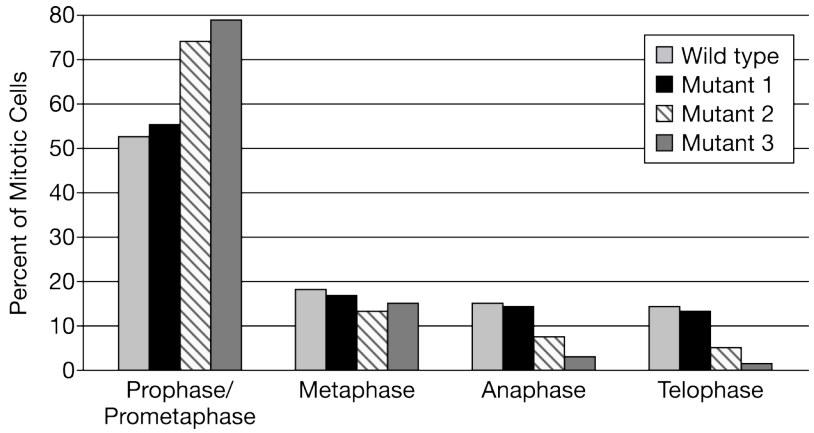


Figure 1. Percent of cells in phases of mitosis

Which of the following patterns is shown by the data?

(A) Mutant 1 cells are more similar to mutant 3 cells than to wild-type cells.

(B) In wild-type cells, the percent of cells in anaphase is twice the amount of those in telophase

(C) In mutant 3 cells, more time is spent in prophase/prometaphase than in the later stages of mitosis. ✓

(D) The percent of mutant 2 cells in anaphase is higher than that of mutant 1 cells.

Answer C

Correct. According the data, mutant 3 had the highest percent of cells in prophase/prometaphase of all cell types and the least time in anaphase and telophase.

34. What is the expected percent change in the DNA content of a typical eukaryotic cell as it progresses through the cell cycle from the start of the G_1 phase to the end of the G_2 phase?

(A) -100%

(B) -50%

(C) $+50\%$

(D) $+100\%$ ✓

4 Cell Communication and Cell Cycle MCQ

Answer D

Correct. The DNA of a typical eukaryotic cell is replicated in the S phase of the cell cycle, which occurs between the G_1 and G_2 phases. DNA replication typically results in a 100% increase in the DNA content of a cell.

35. If chemical signals in the cytoplasm control the progression of a cell to the M phase of the cell cycle, then fusion of a cell in G_1 with a cell in early M phase would most likely result in the

(A) replication of chromosomes only in the G_1 cell

(B) exiting of both cells from the cell cycle and into the G_0 phase

(C) condensation of chromatin in preparation of nuclear division in both cells ✓

(D) transfer of organelles from the G_1 cell to the cell in the M phase

36. Figure 1 shows the number of chromosomes observed in an actively dividing human cell at each stage of cell division.

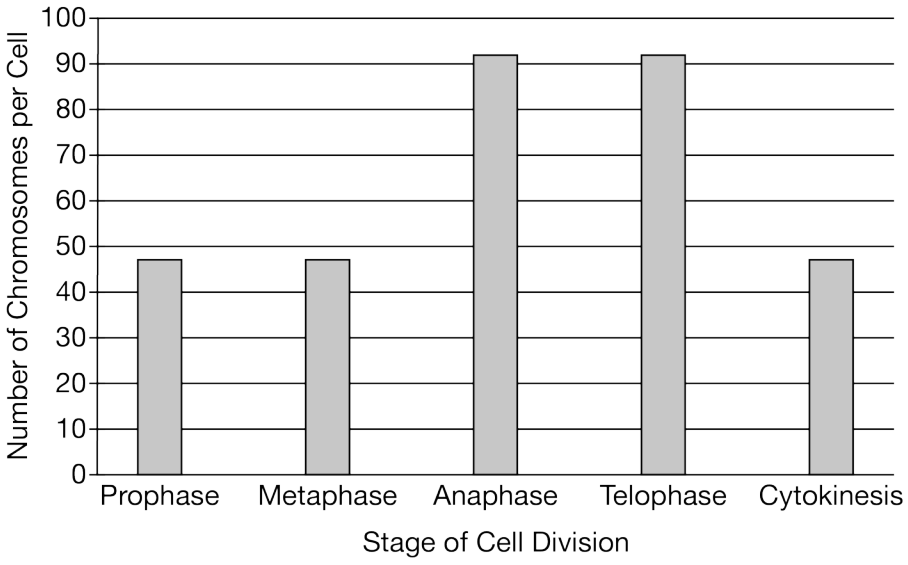


Figure 1. Number of chromosomes in a human cell at different stages of cell division

Which of the following presents a correct interpretation of the changes in chromosome number depicted in Figure 1 ?

4 Cell Communication and Cell Cycle MCQ

(A) DNA replication occurs between metaphase and anaphase, doubling the number of chromosomes. Between telophase and cytokinesis, the cell divides in two, with each cell receiving half of the replicated chromosomes.

(B) New chromosomes formed during prophase are doubled during anaphase and are recombined before cytokinesis.

(C) Chromosomes enter metaphase containing two chromatids attached by a centromere. During anaphase, the chromatids are separated, each becoming a chromosome. Cytokinesis distributes the chromosomes into two separate cells. ✓

(D) At anaphase a cell contains two identical copies of each chromosome, but following telophase, one of the copies is broken down into nucleotides.

Answer C

Correct. This statement accurately describes changes in the chromosome number during the different stages of cell division as represented in the graph. During prophase and metaphase, each chromosome is composed of two chromatids. During anaphase, the two chromatids separate to the opposite ends of the cell, doubling the number of chromosomes in the cell. The number of DNA strands in the cell has not changed, only how they are organized. During cytokinesis, the cell is divided into two daughter cells, each containing half of the chromosomes found in the parent cell during anaphase and telophase.

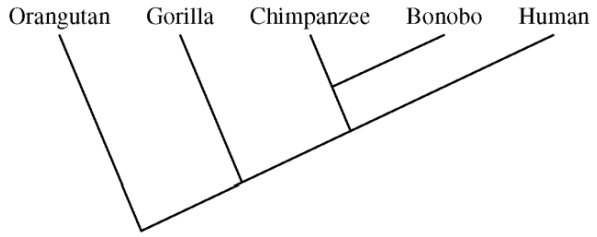
4 Cell Communication and Cell Cycle MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



37. The TAS2R38 receptor protein has been detected on the surface of cells from individuals who are homozygous for the nontaster allele of the *TAS2R38* gene. Which of the following is the most likely effect of the mutations associated with the nontaster allele on *TAS2R38* gene expression?

(A) The mutations change the primary structure of the encoded receptor protein. ✓

(B) The mutations increase the stability of the *TAS2R38* mRNA.

(C) The mutations prevent transcription of the *TAS2R38* gene.

(D) The mutations prevent translation of the *TAS2R38* mRNA.

4 Cell Communication and Cell Cycle MCQ

38. Researchers grew seedlings of corn, *Zea mays*, in loose and compact sand. The researchers measured the amount of time required for the cells in the growing root tips of the seedlings to double in number. The mean cell doubling times for the two groups of seedlings are shown in Figure 1.

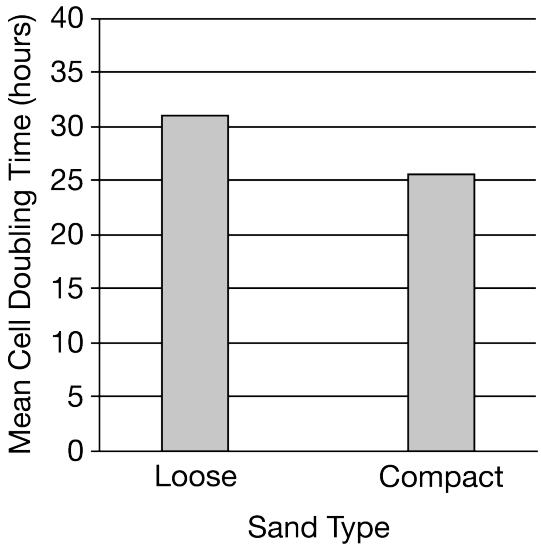


Figure 1. Mean cell doubling times for the growing root tips of *Zea mays* seedlings planted in loose or compact sand

Based on the sample means, which of the following conclusions about the cells in the growing root tips of *Zea mays* seedlings is best supported by the results of the experiment?

(A) The cells of the root tips grow to larger sizes when the seedlings are planted in compact sand than when the seedlings are planted in loose sand.

(B) The average rate of mitotic cell division is greater for the root tips growing in loose sand than for the root tips growing in compact sand.

(C) The average cell cycle time is greater for the root tips growing in compact sand than for the root tips growing in loose sand.

(D) More cells are produced per unit of time in the root tips growing in compact sand than in the root tips growing in loose sand. ✓

Answer D

Correct. According to the figure, the mean cell doubling time of *Zea mays* seedling root tips in compact sand is less than in loose sand. A lower mean cell doubling time indicates an increased rate of mitotic cell division for cells of root tips in compact sand.

4 Cell Communication and Cell Cycle MCQ

39. Cortisol is a hormone produced in response to stress, including starvation, in humans. Which of the following is most likely an immediate effect of a starvation-induced increase in cortisol secretion?

(A) Increased activation of the immune system

(B) Increased urine production by the kidneys

(C) Increased bone and collagen formation

(D) Increased mobilization of fatty acids from fat cells ✓

40. Damaged tissue releases chemicals that activate platelets and stimulate the formation of blood clots. Which of the following statements about the activity of platelets best explains a positive feedback mechanism?

(A) Activated platelets release chemicals that inhibit blood clot formation.

(B) Activated platelets release signaling molecules that inhibit cell division in damaged tissue.

(C) Activated platelets constrict the blood vessels, stopping blood flow.

(D) Activated platelets release chemicals that activate more platelets. ✓

41. Researchers tracked the amount of DNA (measured in picograms) over time beginning with a single cell and continuing through several rounds of cell division. The researchers observed threadlike chromosomes prior to cell division. The threadlike chromosomes disappeared from view shortly after each division. The amount of DNA in picograms per cell over several rounds of cell division is shown in Figure 1.

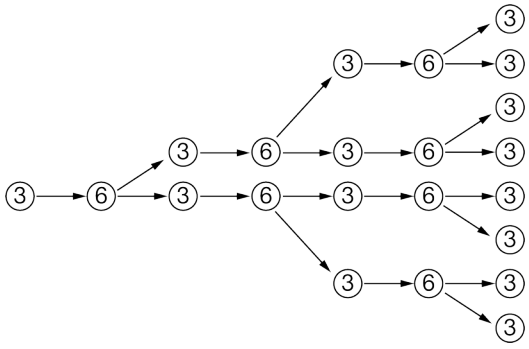


Figure 1. Amount of DNA in picograms per cell over several rounds of cell division

Which of the following statements is consistent with the data in Figure 1?

(A) The cells have a haploid chromosome number of 3.

(B) The cells have a diploid chromosome number of 6.

(C) There is a change from 3 to 6 picograms of DNA because DNA is replicated before each round of cell division. ✓

(D) There is a change from 6 to 3 picograms of DNA after each cell division because the chromosomes lengthen following cell division.

4 Cell Communication and Cell Cycle MCQ

Answer C

Correct. The change from 3 to 6 picograms of DNA prior to replication indicates that the DNA has been replicated. The replicated DNA is condensed into chromosomes during cell division or mitosis. This ensures that the complete genome from a parent cell is passed on to two genetically identical daughter cells.

42. DNA damage can lead to the release of the protein cytochrome *c* from mitochondria. Cytochrome *c* then activates the enzyme caspase-9, resulting in the release of more cytochrome *c*, followed by apoptosis.

Which of the following best explains how the release of cytochrome *c* from mitochondria is regulated by feedback?

(A) It is positive feedback because the release of cytochrome *c* from mitochondria contributes to the production of more mitochondria.

(B) It is negative feedback because the release of cytochrome *c* from mitochondria results in an increase in cell death.

(C) It is positive feedback because the release of cytochrome *c* from mitochondria contributes to the release of more cytochrome. ✓

(D) It is negative feedback because the release of cytochrome *c* from mitochondria is caused by DNA damage.

Answer C

Correct. The release of cytochrome *c* results in the activation of caspase-9, which results in the production of more cytochrome *c*. This is an example of a positive feedback loop.

43. DNA replication occurs

(A) during the S phase of the cell cycle ✓

(B) as the nuclear envelope breaks down in early mitosis

(C) during mitosis but not during meiosis

(D) in animal cells but not in plant cells

(E) only in cells destined to become gametes

4 Cell Communication and Cell Cycle MCQ

44. Human epidermal growth factor (EGF) is a protein that stimulates cell division and growth by binding to its receptor, human epidermal growth factor receptor 2 (HER 2). Certain types of cancer are caused when the overexpression of EGF activates a cell-signaling pathway, leading to uncontrolled cell division.

Which of the following best explains how a drug can be used to inhibit tumor growth caused by EGF overexpression?

(A) The drug binds to HER 2, preventing the activation of the signaling pathway. ✓

(B) The drug acts as a positive regulator of transcription of the gene encoding EGF.

(C) The drug activates the receptor in the absence of EGF, leading to the activation of the signaling pathway.

(D) The drug enables the cells to progress through the S/G_2 checkpoint.

Answer A

Correct. If activating the HER 2 pathway leads to cell division, then an inhibitor that binds to the receptor and blocks its activation, probably by blocking the binding of EGF to HER 2, will block cell division.

45. During mitosis, which of the following normally occurs?

(A) Homologous chromosomes pair.

(B) Replicated chromosomes line up on the equatorial plate. ✓

(C) Tetrads form.

(D) Unreplicated chromosomes become oriented in the center of the cell.

(E) Maternal and paternal chromatids pair.

4 Cell Communication and Cell Cycle MCQ

46. **TABLE 1. DISTRIBUTION OF CONTROL AND EXPERIMENTAL CELLS IN DIFFERENT STAGES OF THE CELL CYCLE AFTER 24 HOURS**

	G_1	S	G_2	M
Control	96	59	48	81
Experimental	142	142	0	0

Researchers claimed that cells treated with radiation are unable to progress through the cell cycle. They tested their claim by exposing a group of cells to radiation (experimental group). A group of unexposed cells served as the control group. After 24 hours, they determined the number of cells in each stage of the cell cycle and confirmed that all differences were statistically significant.

Based on Table 1, which of the following best supports the researchers' claim?

(A) In the experimental group, no cells were able to enter G_2. ✓

(B) In the control group, the number of cells in G_2 was less than the number of cells in M.

(C) In the control group, the number of cells in each stage decreased as the cell cycle continued.

(D) In the experimental group, the number of cells in G_1 was the same as the number of cells in S.

Answer A

Correct. The lack of cells in the G_2 phase supports the researchers' claim that radiation-treated cells are unable to progress through the cell cycle.

47. The drug 5-fluorouracil inhibits thymine production in eukaryotic cells. Which of the following cell cycle stages will be most directly affected by 5-fluorouracil?

(A) The first growth phase (G_1)

(B) Synthesis of DNA phase (S) ✓

(C) Preparation for mitosis (G_2)

(D) Cytokinesis

Answer B

Correct. DNA synthesis that occurs during S requires thymine−containing nucleotides, so inhibition of thymine production will inhibit DNA synthesis.

4 Cell Communication and Cell Cycle MCQ

48. The epidermal growth factor receptor EGFR is a cell surface receptor. When a growth factor binds to EGFR, the receptor is activated. The activated EGFR triggers a signal transduction pathway, which leads to increased frequency of cell division.

Which of the following best predicts the effect of a mutation that causes EGFR to be active in the absence of a growth factor?

(A) Increased apoptosis will lead to abnormal growth of the tissue.

(B) Increased cell division will lead to the formation of a tumor. ✓

(C) Cells will exit the cell cycle, entering a non-dividing G_0 phase.

(D) Fewer cells will be in any of the stages of mitosis.

Answer B

Correct. Growth factors are important signaling molecules that trigger cells to divide at the correct rate and correct time. An EGFR that is active when a growth factor is absent would lead to increased cell division. Uncontrolled cell growth will likely result in a tumor.

49. The endocrine system incorporates feedback mechanisms that maintain homeostasis. Which of the following demonstrates negative feedback by the endocrine system?

(A) During labor, the fetus exerts pressure on the uterine wall, inducing the production of oxytocin, which stimulates uterine wall contraction. The contractions cause the fetus to further push on the wall, increasing the production of oxytocin.

(B) After a meal, blood glucose levels become elevated, stimulating beta cells of the pancreas to release insulin into the blood. Excess glucose is then converted to glycogen in the liver, reducing blood glucose levels. ✓

(C) At high elevation, atmospheric oxygen is more scarce. In response to signals that oxygen is low, the brain decreases an individual's rate of respiration to compensate for the difference.

(D) A transcription factor binds to the regulatory region of a gene, blocking the binding of another transcription factor required for expression.

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4 Cell Communication and Cell Cycle MCQ

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(C) At high elevation, atmospheric oxygen is more scarce. In response to signals that oxygen is low, the brain decreases an individual's rate of respiration to compensate for the difference.

(D) A transcription factor binds to the regulatory region of a gene, blocking the binding of another transcription factor required for expression.

Answer B

This option is correct. The decrease of blood glucose levels as a result of insulin production is an example of how the endocrine system is involved in negative feedback, allowing organisms to maintain their internal environments.

51. During a fight-or-flight response, epinephrine is released into the body's circulatory system and transported throughout the body. Some cells exhibit a response to the epinephrine while other cells do not.

Which of the following justifies the claim that differences in components of cell signaling pathways explain the different responses to epinephrine?

(A) Cell signaling depends on the ability to detect a signal molecule. Not all cells have receptors for epinephrine. Only cells with such receptors are capable of responding. ✓

(B) Cell signaling depends on the transduction of a received signal by the nervous system. Not all cells are close enough to a synapse to receive the signal and respond.

(C) Cell signaling depends on the signal being able to diffuse through the cell membrane. Epinephrine is incapable of diffusing through some plasma membranes because of the membrane's phospholipid composition.

(D) Cell signaling requires reception, transduction, and response. All cells can receive epinephrine, all cells respond with a pathway, but only select cells have the proper coding in their DNA to respond.

Answer A

Correct. Cell signaling starts with the reception of the signal. Not all cells are equipped to receive all signals. Only those cells with the correct receptor to a signal will be able to respond to a signal such as epinephrine.

4 Cell Communication and Cell Cycle MCQ

The epinephrine signaling pathway plays a role in regulating glucose homeostasis in muscle cells. The signaling pathway is activated by the binding of epinephrine to the beta-2 adrenergic receptor. A simplified model of the epinephrine signaling pathway is represented in Figure 1.

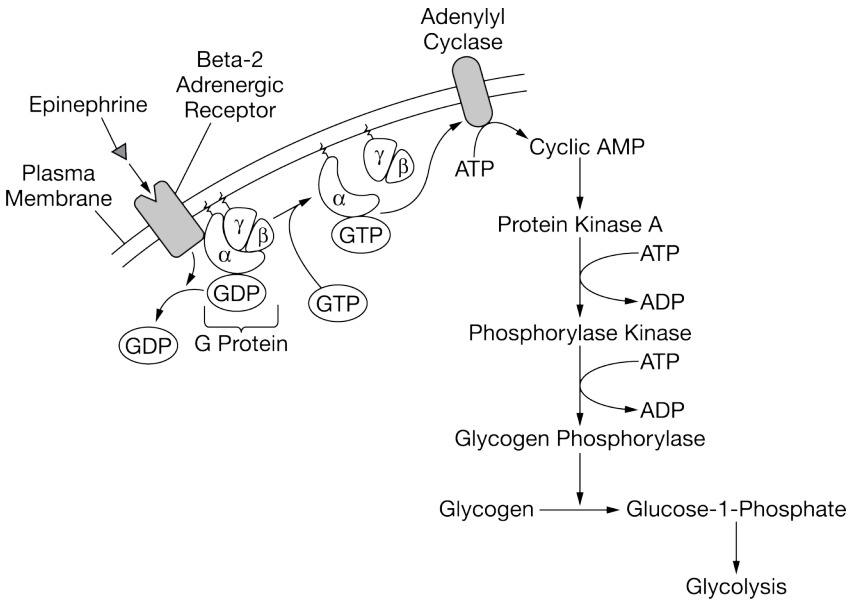


Figure 1. A simplified model of the epinephrine signaling pathway in muscle cells

52. Which of the following outcomes will most likely result from the irreversible binding of GDP to the G protein?

(A) The intracellular concentration of glycogen will increase. ✓

(B) The intracellular concentration of activated protein kinase A will increase.

(C) The intracellular concentration of cyclic AMP will increase.

(D) The intracellular concentration of glucose-1-phosphate will increase.

Answer A

Correct. Activation of the G protein requires the exchange of GDP for GTP. The irreversible binding of GDP to the G protein will block activation of the G protein, which will prevent epinephrine stimulation of glycogen breakdown. Consequently, the intracellular concentration of glycogen will increase.

53. A researcher claims that the epinephrine signaling pathway controls a catabolic process in muscle cells. Which of the following statements best helps justify the researcher's claim?

4 Cell Communication and Cell Cycle MCQ

(A) Epinephrine is a signaling molecule that binds to a transmembrane protein.
(B) The G protein in the epinephrine signaling pathway consists of three different subunits.
(C) Phosphorylase kinase catalyzes the hydrolysis of ATP.
(D) Glycogen phosphorylase catalyzes the conversion of glycogen to glucose-1-phosphate. ✓

Answer D

Correct. Glycogen phosphorylase catalyzes the breakdown of glycogen, which is a catabolic process. Consequently, the statement helps justify the researcher's claim.

54. Cyclic AMP phosphodiesterase is an enzyme that catalyzes the conversion of cyclic AMP to a different molecule. Which of the following best predicts the effect of inhibiting cyclic AMP phosphodiesterase in a muscle cell stimulated by epinephrine?

(A) The concentration of cyclic AMP will decrease because adenylyl cyclase will no longer be activated.
(B) The G protein will diffuse out of the cell because it will no longer bind to the plasma membrane.
(C) Phosphorylase kinase will remain active because protein kinase A will no longer be deactivated. ✓
(D) Glycolysis will stop because epinephrine signaling will no longer stimulate glycogen breakdown.

Answer C

Correct. Phosphorylase kinase is activated by protein kinase A, which is activated by binding to cyclic AMP. Cyclic AMP phosphodiesterase causes the deactivation of protein kinase A by lowering the intracellular levels of cyclic AMP. Inhibiting cyclic AMP phosphodiesterase will result in protein kinase A remaining active, which will result in phosphorylase kinase remaining active.

55. Which of the following statements best describes the role of adenylyl cyclase in the epinephrine signaling pathway?

(A) It converts a polymer to its monomer subunits.
(B) It moves substances across the plasma membrane.
(C) It accelerates the production of a second messenger. ✓
(D) It transfers phosphate groups from ATP to protein substrates.

4 Cell Communication and Cell Cycle MCQ

Answer C

Correct. When activated, adenylyl cyclase accelerates a reaction in which ATP is converted to cyclic AMP, which acts as a second messenger in the epinephrine signaling pathway.

56. Based on Figure 1, which of the following statements best describes the epinephrine signaling pathway?

(A) It involves the opening and closing of ion channels.

(B) In involves enzymes activating other enzymes. ✓

(C) It involves changes in the expression of target genes.

(D) It involves protons moving down a concentration gradient.

Answer B

Correct. Based on Figure 1, the epinephrine signaling pathway involves enzymes activating other enzymes. For example, the pathway includes several protein kinases, enzymes that catalyze the transfer of a phosphate group from ATP to a protein substrate. As represented in Figure 1, protein kinase A catalyzes the transfer of a phosphate group from ATP to phosphorylase kinase, which results in the activation of phosphorylase kinase. The activated phosphorylase kinase activates glycogen phosphorylase in a similar manner.

4 Cell Communication and Cell Cycle MCQ

57. Figure 1 is a proposed model of the feedback system controlling erythrocyte (red blood cell) production.

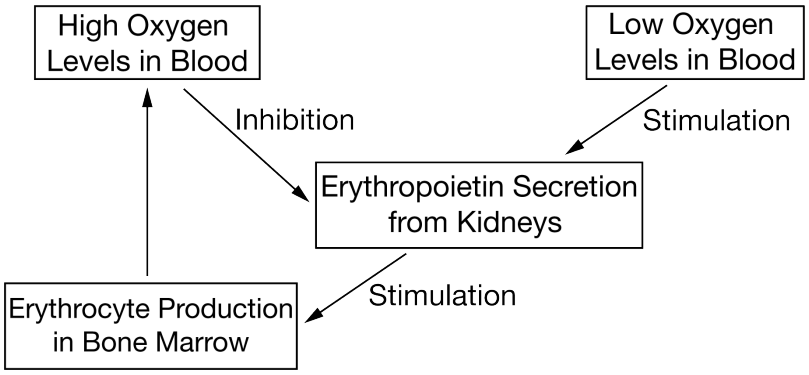


Figure 1. Model of erythrocyte production control

Air is less dense at very high elevations, so less oxygen is available than in the denser air at sea level. Based on the model in Figure 1, if a person travels from sea level to a high elevation location, which of the following correctly predicts the response to the decreased blood oxygen level?

(A) More erythropoietin will be secreted from the kidneys, decreasing production of erythrocytes.

(B) More erythropoietin will be secreted from the kidneys, increasing production of erythrocytes. ✓

(C) Less erythropoietin will be secreted from the kidneys, decreasing production of erythrocytes.

(D) Less erythropoietin will be secreted from the kidneys, increasing production of erythrocytes.

Answer B

Correct. Erythropoietin production is triggered by a negative feedback mechanism. When kidney cells detect low oxygen levels, the kidney secretes more erythropoietin, which triggers the bone marrow to increase erythrocyte production.

58. Ethylene causes fruits to ripen. In a signaling pathway, receptors activate transcription factors, which ultimately leads to ripening.

Which of the following best supports the claim that ethylene initiates the signal transduction pathway that leads to ripening of fruit?

4 Cell Communication and Cell Cycle MCQ

(A) Ethylene is a simple gaseous molecule, which makes it easily detected by receptors.

(B) Fruit will ripen in closed containers without exposure to air.

(C) Ethylene synthesis is under both positive and negative feedback regulation.

(D) Loss-of-function mutations in ethylene receptors result in changes to the ripening process. ✓

Answer D

Correct. If ethylene is not perceived by the receptors, then ripening will not occur. This best justifies the claim that ethylene initiates the signal transduction pathway.

59. Platelets are fragments of larger cells and normally circulate in the blood without adhering to blood vessel walls. When the wall of a blood vessel is damaged, collagen fibers in the wall are exposed to the interior of the blood vessel. The exposed fibers and chemicals released from the endothelial cells that line the blood vessel attract platelets, which start to form a plug and release other chemicals (Figure 1).

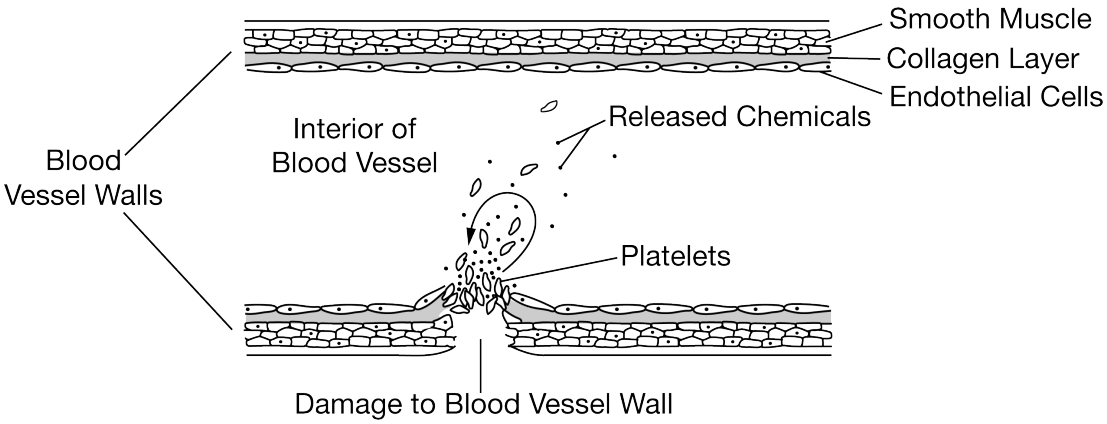


Figure 1. Formation of a platelet plug in a damaged blood vessel wall

Which of the following best explains the feedback mechanism illustrated in Figure 1 ?

(A) This is an example of positive feedback, because the few platelets that initially bind attract more platelets to the damaged area. ✓

(B) This is an example of positive feedback, because it results from the interactions among collagen, endothelial cells, and platelets.

(C) This is an example of negative feedback, because a large clump of platelets can block the blood vessel and prevent blood flow through it.

(D) This is an example of negative feedback, because the accumulation of platelets returns the open blood vessel wall to a closed state.

4 Cell Communication and Cell Cycle MCQ

60. Beta cells, a type of cell found in the pancreas, are responsible for producing, storing, and releasing the hormone (chemical signal) insulin. An increase in blood glucose levels stimulates beta cells to produce insulin, which facilitates the uptake of glucose by body cells and reduces the glucose levels in the blood. As the blood glucose levels decrease, beta cells are no longer stimulated and stop producing insulin.

Which of the following best explains the feedback mechanism that regulates blood glucose levels?

(A) Regulation of blood glucose levels demonstrates positive feedback, because as the blood glucose levels increase, insulin acts to raise blood glucose levels.

(B) Regulation of blood glucose levels demonstrates negative feedback, because as the blood glucose levels increase, insulin acts to lower blood glucose levels. ✓

(C) Regulation of blood glucose levels demonstrates positive feedback, because as the production of insulin by beta cells increases, blood glucose levels increase.

(D) Regulation of blood glucose levels demonstrates negative feedback, because as the production of insulin by beta cells decreases, blood glucose levels decrease.

Answer B

Correct. A negative feedback loop occurs when a signal is detected and causes a decrease in production. As more insulin is released, blood glucose levels decrease, which in turn decreases insulin production.

4 Cell Communication and Cell Cycle MCQ

61. Fibroblast growth factor receptors (FGFRs) are transmembrane proteins that regulate cellular processes such as cell proliferation and differentiation. The extracellular domains of FGFR proteins bind specifically to signaling molecules called fibroblast growth factors. The intracellular domains of FGFR proteins function as protein kinases, enzymes that transfer phosphate groups from ATP to protein substrates.

FGFR activation occurs when binding by fibroblast growth factors causes FGFR proteins in the plasma membrane to become closely associated with each other. The association of two FGFR proteins stimulates protein kinase activity, which triggers the activation of intracellular signaling pathways. A simplified model of FGFR activation is represented in Figure 1.

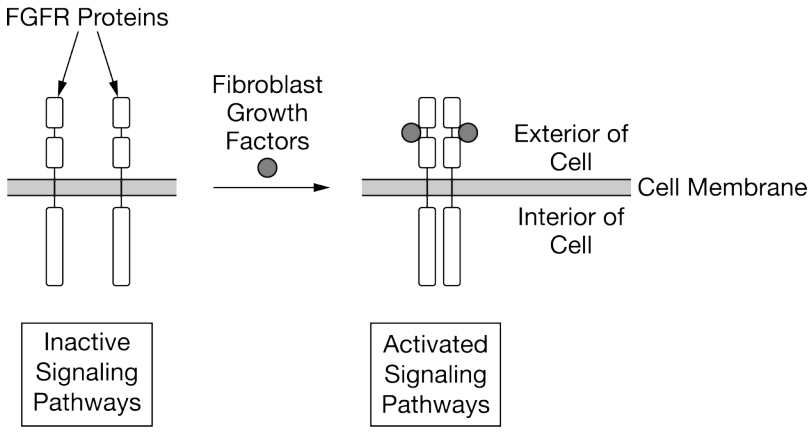


Figure 1. A simplified model of FGFR activation

Which of the following changes in the FGFR signaling pathway is most likely to result in uncontrolled cell proliferation?

(A) The irreversible association of FGFR proteins ✓

(B) The loss of the FGFR protein kinase function

(C) A decrease in the intracellular concentration of ATP

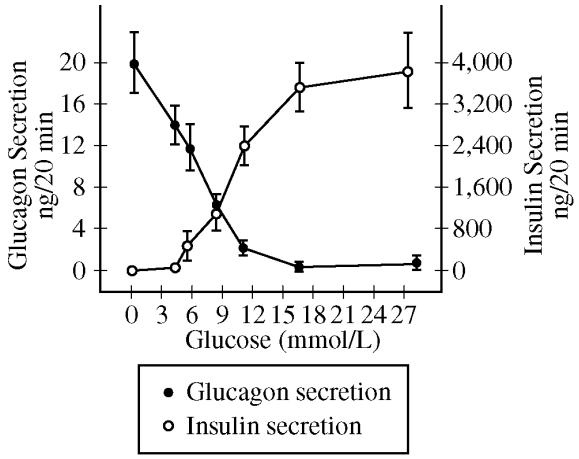
(D) A decrease in the extracellular concentrations of fibroblast growth factors

Answer A

Correct. The irreversible association of FGFR proteins will most likely result in the permanent activation of the FGFR signaling pathway. Because the FGFR signaling pathway regulates cell proliferation, the irreversible association of FGFR proteins will result in uncontrolled cell proliferation.

4 Cell Communication and Cell Cycle MCQ

62.



The graph above shows changes in glucagon and insulin secretions at different concentrations of blood glucose. Which of the following feedback mechanisms is best supported by the data?

(A) A falling glucagon level causes a rise in the insulin level, which maintains equal amounts of both hormones in the blood.

(B) A high glucagon level causes a rise in the insulin level, which maintains high levels of both hormones in the blood.

(C) A low glucose level causes the release of glucagon, which stimulates the release of more glucose from tissues, which in turn lowers the amount of glucagon being released. ✓

(D) A low glucose level causes the release of insulin, which stimulates the release of more glucose from tissues, which in turn increases the amount of insulin being released.

4 Cell Communication and Cell Cycle MCQ

63. Glucocorticoids are steroid hormones that control cellular responses through several different signaling pathways. One of the signaling pathways involves the glucocorticoid receptor, an intracellular protein that is activated by binding to a glucocorticoid molecule. A simplified model of the glucocorticoid receptor signaling pathway is represented in Figure 1.

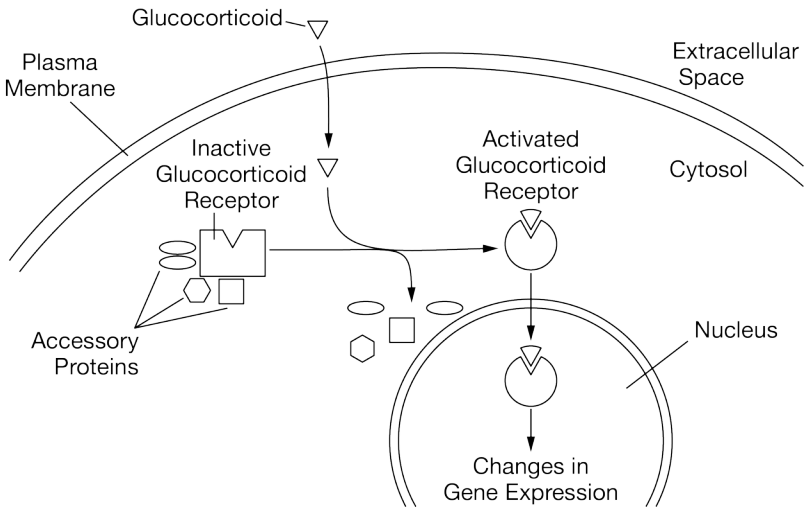


Figure 1. A simplified model of the glucocorticoid receptor signaling pathway

Which of the following statements best predicts the effect of a mutation that results in a loss of the glucocorticoid receptor's ligand binding function?

(A) The transduction of the glucocorticoid signal across the plasma membrane will be blocked.

(B) The glucocorticoid receptor will remain associated with the accessory proteins. ✓

(C) The rate of diffusion of glucocorticoid molecules into the cell will increase.

(D) The concentration of glucocorticoid receptors inside the nucleus will increase.

Answer B

Correct. In the absence of glucocorticoid signaling, the glucocorticoid receptor is maintained in its inactive state in the cytosol by its association with accessory proteins. The binding of a glucocorticoid molecule to the glucocorticoid receptor induces a conformational change in the glucocorticoid receptor that results in its dissociation from the accessory proteins and translocation to the nucleus. A mutation that results in a loss of the glucocorticoid receptor's ligand binding function will result in the glucocorticoid receptor remaining associated with the accessory proteins in a multiprotein complex in the cytosol.

4 Cell Communication and Cell Cycle MCQ

64. Glycogen synthetase kinase 3 beta is a protein kinase that has been implicated in many types of cancer. Depending on the cell type, the gene for glycogen synthetase kinase 3 beta $(GSK3\beta)$ can act either as an oncogene or as a tumor suppressor.

Which of the following best predicts how $GSK3\beta$ mutations can lead to the development of cancer?

(A) Cells with inactive $GSK3\beta$ fail to trigger apoptosis. ✓

(B) Cells with inactive $GSK3\beta$ fail to proceed past the $\mathrm{G_2/M}$ checkpoint.

(C) Cells with overactive $GSK3\beta$ are more likely to repair DNA damage.

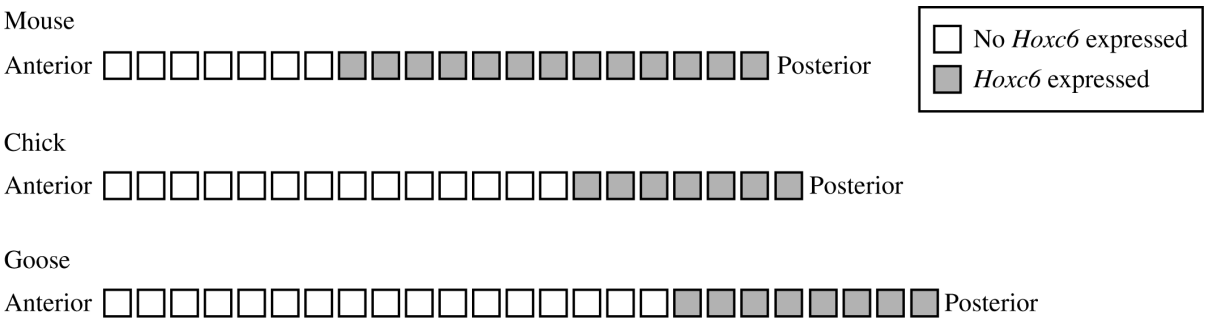
(D) Cells with overactive $GSK3\beta$ have longer cell cycles.

Answer A

Correct. If apoptosis cannot be initiated in cells with $GSK3\beta$ mutations, those cells are more likely to proliferate, which could lead to cancer.

65. In animals, the *hox* genes encode a family of transcription factors that are important for proper development of embryonic segments and are widely conserved in organisms. The figure below shows the embryonic segments in which one such gene, *Hoxc6*, is expressed in the embryo of a mouse, a chick, and a goose. Embryonic segments are counted from the anterior end.

During the formation of vertebrae, the most anterior embryonic segment that expresses *Hoxc6* marks the end of the cervical (neck) vertebrae and the beginning of the thoracic (rib) vertebrae. All mammals have seven cervical vertebrae.



Which of the following statements is most likely to be true?

(A) The chick and the goose have the same number of thoracic vertebrae.

(B) The most anterior expression of *Hoxc6* is the eighth vertebra in mammals. ✓

(C) *Hoxc6* is expressed in the same embryonic segments in birds and mammals.

(D) *Hoxc6* is expressed in the same vertebra at the anterior end of all bird embryos.

4 Cell Communication and Cell Cycle MCQ

66. The insulin receptor is a transmembrane protein that plays a role in the regulation of glucose homeostasis. The receptor's extracellular domain binds specifically to the peptide hormone insulin. The receptor's intracellular domain interacts with cellular factors. The binding of insulin to the receptor stimulates a signal transduction pathway that results in the subcellular translocation of GLUT4, a glucose transport protein that is stored in vesicles inside the cell. A simplified model of the insulin receptor–signaling pathway is shown in Figure 1.

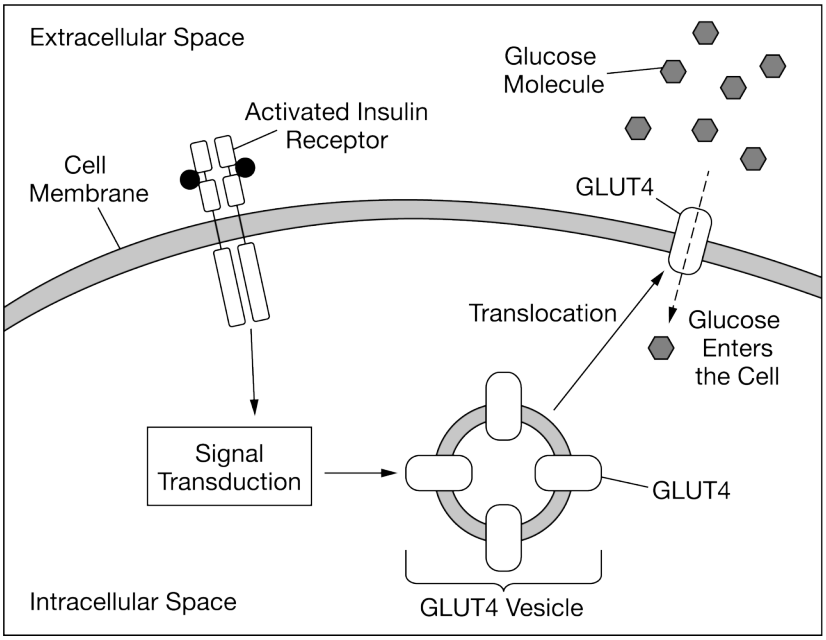


Figure 1. A simplified model of the insulin receptor–signaling pathway

Which of the following statements best predicts the effect of a loss of function of the insulin receptor's intracellular domain?

(A) The stimulation of the signal transduction pathway will increase.

(B) The storage of GLUT4 in vesicles inside the cell will increase. ✓

(C) The number of GLUT4 molecules in the plasma membrane will increase.

(D) The concentration of glucose inside the cell will increase.

Answer B

Correct. The insulin receptor's intracellular domain is required for the stimulation of the signal transduction pathway. If the signal transduction pathway is not stimulated by the insulin receptor, the GLUT4 vesicles will not be translocated to the plasma membrane. Consequently, the storage of GLUT4 in vesicles inside the cell will increase.

4 Cell Communication and Cell Cycle MCQ

67. Signal transduction may result in changes in gene expression and cell function, which may alter phenotype in an embryo. An example is the expression of the SRY gene, which triggers the male sexual development pathway in mammals. This gene is found on the Y chromosome.

Which statement provides the evidence to justify the claim that signal transduction may result in an altered phenotype?

(A) If the SRY gene is absent or nonfunctional, the embryo will exhibit male sexual development.

(B) If the SRY gene is absent or nonfunctional, the embryo will exhibit female sexual development. ✓

(C) An embryo with a male sex chromosome will always exhibit male sexual development.

(D) An embryo with two male sex chromosomes will always exhibit male sexual development.

Answer B

Correct. The SRY gene must be present and functional for normal male sexual development. It is the proper functioning of the SRY gene that is important, and this is the best statement to support the claim.

68. Two types of cholesterol transport proteins, low-density lipoproteins (LDL) and high-density lipoproteins (HDL), bind to cholesterol and carry it through the bloodstream. Familial hypercholesterolemia (FH) is characterized by high cholesterol levels in the blood, which can lead to cardiovascular disease.

FH is associated with a loss-of-function mutation of a gene that encodes LDL receptors in liver cells. Individuals who are heterozygous produce lower-than-normal amounts of the LDL receptors, and individuals who are homozygous for the mutant allele have no LDL receptor function.

Individuals with FH can be treated with drugs that result in increased production of LDL receptors in liver cells. Which of the following best explains the observation that the drugs can effectively control blood cholesterol levels in individuals who are heterozygous but are not effective in individuals homozygous for the mutant allele?

(A) The drugs repair the mutant allele by copying the wild-type allele.

(B) The drugs prevent cholesterol from entering the liver cells in individuals who are heterozygous but not in individuals who are homozygous for the mutant allele.

(C) Cholesterol molecules primarily bind to HDL receptors in individuals with FH.

(D) There must be at least one copy of the wild-type LDL receptor allele to produce functional LDL receptors. ✓

69. The mechanism of action of many common medications involves interfering with the normal pathways that cells use to respond to hormone signals. Which of the following best describes a drug interaction that directly interferes with a signal transduction pathway?

4 Cell Communication and Cell Cycle MCQ

(A) A medication causes the cell to absorb more of a particular mineral, eventually poisoning the cell.

(B) A medication enters the target cell and inhibits an enzyme that normally synthesizes a second messenger. ✓

(C) A medication enters the target cell's nucleus and acts as a mutagen.

(D) A medication interrupts the transcription of ribosomal RNA genes.

Answer B

This option is correct. The student is asked to justify the claim that changes in signal transduction pathways can alter cellular response. Second messengers are involved in signal transduction pathways, and if the second messenger is not present, the pathway is blocked.

70. The relative amounts of DNA present in the nucleus of a cell at four different stages of the life cycle are shown in Figure 1.

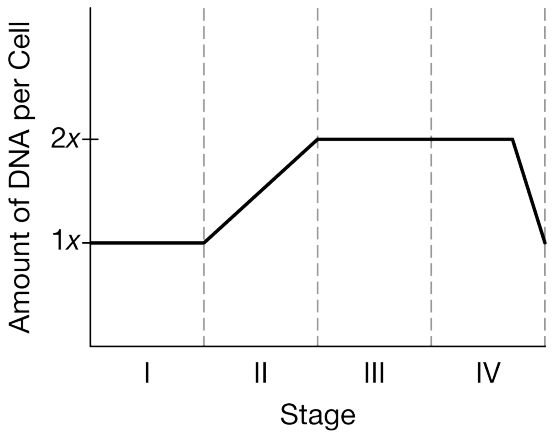


Figure 1. Relative amounts of DNA present in the nucleus of a cell

Based on Figure 1, which of the following statements correctly links a stage of the cell cycle with the event occurring at that stage?

(A) Stage I represents the G_2 phase of the cell cycle.

(B) Synthesis of sufficient DNA for two daughter cells occurs in stage II. ✓

(C) Stage III includes mitosis.

(D) The replication of genetic material occurs in stage IV.

4 Cell Communication and Cell Cycle MCQ

Answer B

Correct. Stage II represents the S (synthesis) phase of interphase. The genetic information (DNA) is doubling. If this did not occur, the daughter cells formed as a result of the cell cycle would have too little genetic information (chromosomes). This process is visible on the graph, as the copies of genetic information increase from one to two copies.

71. Which of the following best describes the role of mitosis in the cell cycle?

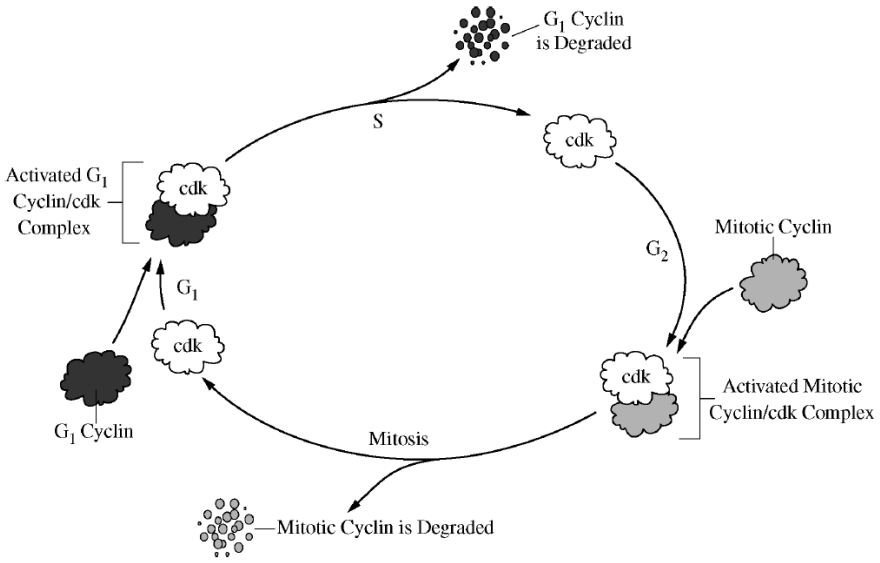
(A) Distributing replicated chromosomes to daughter nuclei ✓

(B) Dividing the cytoplasm to form four gametes

(C) Producing organelles and replicating chromosomes

(D) Exchanging genetic material between homologous chromosomes

72.



Based on the model of eukaryotic cell cycle regulation shown in the figure, which of the following best describes the effect of a drug that blocks the production of the mitotic cyclin?

(A) The cell cycle would proceed uncontrollably, and the cell would become cancerous.

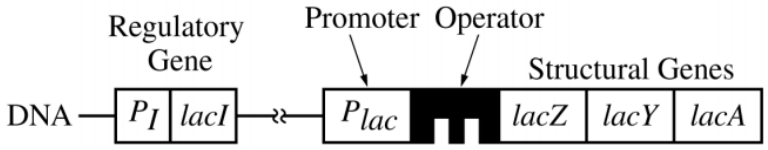
(B) The G1 cyclin would functionally replace mitotic cyclin, and the cell would continue dividing normally.

(C) DNA synthesis would be prevented, and the cell would stop dividing.

(D) The cell would be prevented from entering mitosis, and the cell would stop dividing. ✓

4 Cell Communication and Cell Cycle MCQ

LAC OPERON STRUCTURE



The functions of the loci of the *lac* operon shown in the diagram are described in the table below.

Locus	Function
P_I	Attachment site for RNA polymerase
lacI	Encodes a repressor protein that prevents transcription of the structural genes of the *lac* operon
P_{lac}	Attachment site for RNA polymerase
Operator	Binding site for the repressor protein
lacZ	Encodes beta-galactosidase, the enzyme that digests lactose to glucose and galactose
lacY	Encodes lactose permease, the channel through which lactose moves into the cell
lacA	Encodes galactoside transacetylase

The diagram above represents a segment of the *E. coli* chromosome that contains the *lacI* gene and part of the *lac* operon, a coordinately regulated set of genes that are required for the metabolism of lactose. The presence of lactose, which causes the repressor to be released from the operator, results in increased transcription of the *lac* operon.

73. Which of the following describes the most likely consequence of a negative feedback pathway involving the *lac* operon?

(A) The breakdown products of lactose block cell division.

(B) Diffusion of lactose across the plasma membrane would stop in the presence of lactose.

(C) Expression of the *lac* operon diminishes as lactose is depleted. ✓

(D) Binding of DNA by the repressor would increase in the presence of lactose.

4 Cell Communication and Cell Cycle MCQ

74. Figure 1 shows a model of a signal transduction cascade, initiated by the binding of a ligand to the transmembrane receptor protein A.

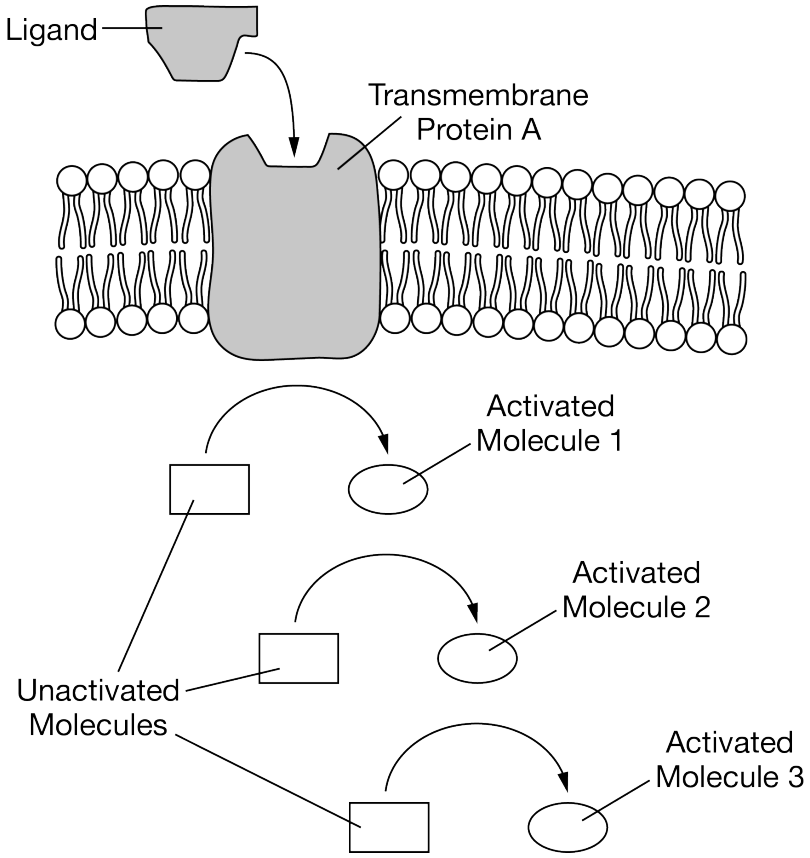


Figure 1. Model of signal transduction cascade incorporating protein A

A DNA mutation changes the shape of the extracellular domain of transmembrane receptor protein A produced by the cell. Which of the following predictions is the most likely consequence of the mutation?

(A) Production of activated molecule 1 will stop, but production of activated molecules 2 and 3 will continue.

(B) The molecule that normally binds to protein A will no longer attach, deactivating the cellular response. ✓

(C) The molecule that normally binds to protein A will not enter the cell, thus no cellular response will occur.

(D) Since protein A is embedded in the membrane, the mutation will be silent and not affect the cellular response.

4 Cell Communication and Cell Cycle MCQ

Answer B

Correct. In this signal transduction model, the molecule does not enter the cell but binds to the extracellular surface receptor of protein A, thus causing a cascade reaction that produces the cellular response. Ligand binding and activation depends upon the ligand, and the receptor have the appropriate complementary shapes. Since the specified mutation changes the shape of the ligand-binding site, the cascade will not be activated, and the cell will not respond to the presence of the ligand.

75. Cancer cells behave differently than normal body cells. For example, they ignore signals that tell them to stop dividing.

Which of the following conditions will most likely cause a normal body cell to become a cancer cell?

(A) The environment already contains cancer cells.

(B) The environment has an abundance of nutrients.

(C) The environment lacks signals that would otherwise tell the cell to stop dividing.

(D) The environment contains mutagens that induce mutations that affect cell-cycle regulator proteins. ✓

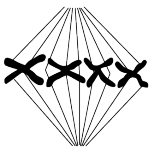
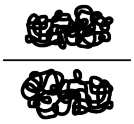
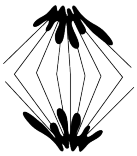
Answer D

Correct. Cancerous cells often produce abnormal forms of cell-cycle regulators, allowing them to divide even when it is not appropriate to do so.

4 Cell Communication and Cell Cycle MCQ

76. A student used microscopy to investigate the relative lengths of the different stages of mitosis. The student prepared slides of cells isolated from a growing onion root tip and viewed the slides under a dissecting microscope. The student then made diagrams of cells that were in different stages of mitosis and counted the number of cells that were in each of those stages. The student's data are presented in Table 1.

Table 1. Number of cells in each of four different stages of mitosis

Stage of Mitosis	Number of Cells
	7
	4
	25
	2

Based on the data, the percent of the mitotic cells that were in metaphase is closest to which of the following?

(A) 5%

(B) 11%

(C) 18% ✓

(D) 66%

Answer C

Correct. The number of cells that were in metaphase equals 7, and the total number of mitotic cells equals 38. The percent of mitotic cells in metaphase is calculated as follows: divide 7 by 38 and multiply

by 100. The correct answer is 18 percent.

77. The tumor suppressor protein p53 binds to DNA and activates target genes, which results in the synthesis of p21, CD95, and other proteins. The p21 protein promotes cell-cycle arrest, whereas the CD95 protein promotes apoptosis.

Which of the following will most likely result from a loss of p53 function?

(A) Rapid cell growth without cell division

(B) Immediate activation of apoptosis pathways

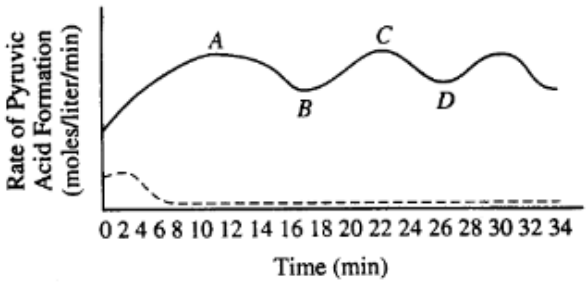
(C) Uncontrolled cell proliferation ✓

(D) Increased expression of p53 target genes

Answer C

Correct. A loss of p53 function will result in unregulated cell proliferation because the p21 protein will not be synthesized. In the absence of the p21 protein, a cell will continue to progress through the cell cycle without stopping.

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.



A tissue culture of vertebrate muscle was provided with a constant excess supply of glucose under anaerobic conditions starting at time zero and the amounts of pyruvic acid and ATP produced were measured. The solid line in the graph above represents the pyruvic acid produced in moles per liter per minute. ATP levels were also found to be highest at points *A* and *C*, lowest at *B* and *D*. A second culture was set up under the same conditions, except that substance **X** was added, and the results are indicated by the dotted line.

78. The rate of pyruvic acid formation fluctuates because

4 Cell Communication and Cell Cycle MCQ

(A) all glucose has reacted

(B) all enzymes have been used up

(C) the reaction is accelerated by positive feedback

(D) the reaction is affected by negative feedback ✓

(E) coenzymes have begun to function

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79. The binding of erythropoietin to the erythropoietin receptor (EPoR) stimulates intracellular signaling in certain cells, which causes them to differentiate to red blood cells. The EPoR is a transmembrane protein with two cytoplasmic domains, each of which is bound by a kinase protein, JAK 2. The binding of erythropoietin to the EPoR causes the cytoplasmic domains of the EPoR to undergo conformational changes that activate the bound JAK 2 kinases. The JAK 2 kinases then phosphorylate several intracellular proteins, including STAT 5, whose resulting activity causes the cells to differentiate to red blood cells.

Which of the following explains how a mutation that removes a portion of each cytoplasmic domain of EPoR most likely affects differentiation of the cells?

(A) The cells will differentiate more rapidly than they typically do because the JAK 2 kinases will always be active.

(B) The cells will mature to another cell type because the JAK 2 kinases will phosphorylate inappropriate cellular proteins.

(C) STAT 5 will not be activated because erythropoietin cannot bind to the EPoR.

(D) STAT 5 will not be activated because it will not be phosphorylated by the JAK 2 kinases. ✓

Answer D

Correct. If a portion of the EPoR cytoplasmic domain is missing, the JAK 2 kinases may be unable to bind to the EPoR cytoplasmic domains. Even if the kinases can bind, the EPoR cytoplasmic domains are unlikely to undergo the normal conformational changes that are required to activate the JAK 2 kinases. In turn, STAT 5 will not be phosphorylated and activated.

80. A student claims that the Y chromosome contains the sex-determining region gene, known as the *SRY* gene, which causes male fetuses to develop testes.

Which of the following provides correct information about cell signaling that supports the claim?

4 Cell Communication and Cell Cycle MCQ

(A) The SRY gene produces a protein that binds to specific regions of DNA in certain tissues, which affects the development of these tissues. ✓

(B) The SRY gene produces a protein that deletes portions of the X chromosome in males so that male characteristics can develop.

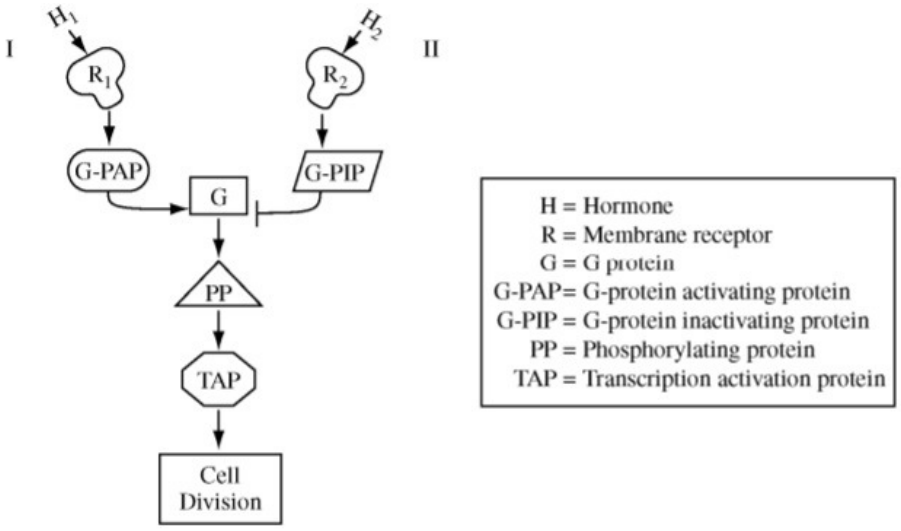
(C) The SRY gene produces an RNA segment that is exported from specific cells and targets the developing gonads.

(D) The SRY gene is found only in tissues of the developing gonads.

Answer A

Correct. SRY activation results in a signal transduction pathway that produces the male phenotype. SRY activation produces transcription factors that bind to specific regions of DNA and results in male sexual development.

81.



The diagram above represents a model of signal transduction pathways (I and II) in a cell that is targeted by two different hormones (H_1 and H_2). The components of the signal transduction pathways are identified in the figure legend. Each cellular molecule in both pathways can exist in an inactive or active form. When the components in pathway I are sequentially activated, the TAP molecules promote cell division. When the components in pathway II are sequentially activated, downstream signaling by the G protein is inhibited. Based on the model, which of the following mutations is most likely to result in a cell that will generate a cancerous tumor?

(A) A mutation in the gene encoding PP that results in a nonfunctional protein

(B) A mutation in the gene encoding G-PIP that results in a nonfunctional protein ✓

(C) A mutation in the gene encoding R_1 so that it is inactive even in the presence of H_1

(D) A mutation in the gene encoding R_2 so that it is active even in the absence of H_2

4 Cell Communication and Cell Cycle MCQ

82. The vertebrate forelimb initially develops in the embryo as a solid mass of tissue. As development progresses, the solid mass near the end of the forelimb is remodeled into individual digits. Which of the following best explains the role of apoptosis in remodeling of the forelimb?

(A) Apoptosis replaces old cells with new ones that are less likely to contain mutations.

(B) Apoptosis involves the regulated activation of proteins in specific cells of the developing forelimb that leads to the death of those cells. ✓

(C) Apoptosis involves the destruction of extra cells in the developing forelimb, which provides nutrients for phagocytic cells.

(D) Apoptosis in the developing forelimb triggers the differentiation of cells whose fate was not already determined.

83. The coagulation cascade controls blood clot formation in response to blood vessel injury. Thrombin is an enzyme that plays a key role in regulating the coagulation cascade. A simplified model of thrombin's role in regulating the coagulation cascade is represented in Figure 1.

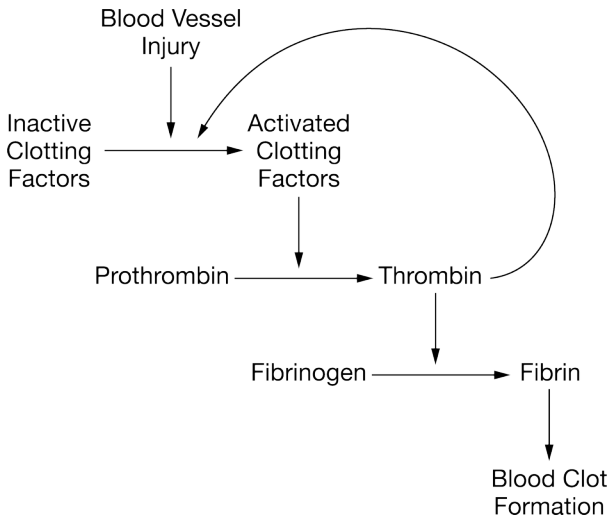


Figure 1. A simplified model of thrombin's role in regulating the coagulation cascade

Argatroban is a competitive inhibitor of thrombin. Which of the following effects on the coagulation cascade is most likely to result from inhibiting thrombin activity with argatroban?

(A) The activation of clotting factors will be blocked.

(B) The rate of fibrin formation will decrease. ✓

(C) Thrombin will be converted to prothrombin.

(D) The rate of blood clot formation will increase.

4 Cell Communication and Cell Cycle MCQ

Answer B

Correct. Thrombin is involved in a positive feedback loop. When activated, thrombin activates clotting factors that convert prothrombin to more thrombin. Activated thrombin also catalyzes the conversion of fibrinogen to fibrin, which is an important step in blood clot formation. Although activated clotting factors will catalyze the conversion of prothrombin to thrombin, inhibiting thrombin activity with argatroban will interfere with the positive feedback loop that generates more active thrombin. A decrease in the rate of thrombin formation will result in a decrease in the rate of fibrin formation.

84.

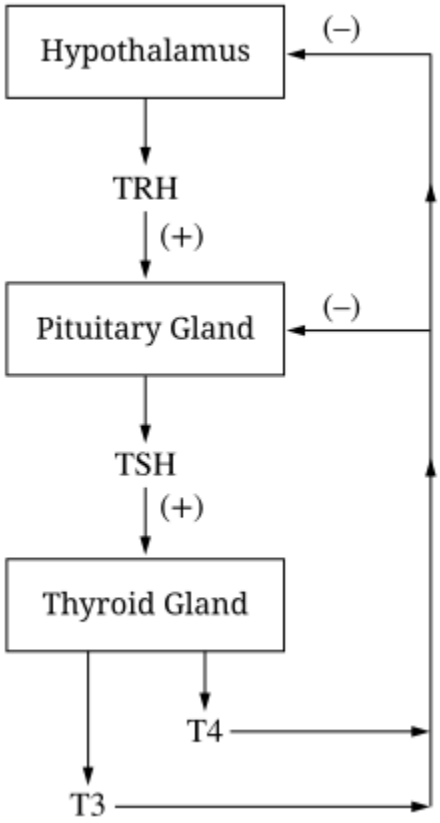


Figure 1. Regulation of hormones secreted by the thyroid gland. The + indicates stimulation and the − indicates inhibition.

Based on Figure 1, which of the following most likely occurs in response to an increase in the secretion of thyroid hormones T3 and T4 ?

4 Cell Communication and Cell Cycle MCQ

(A) The hypothalamus secretes more TRH to maintain the increase in thyroid hormone secretion.
(B) The pituitary gland decreases secretion of TSH, which results in an increase in TRH secretion.
(C) The thyroid hormones directly inhibit the hypothalamus, which results in a decrease in TRH secretion. ✓
(D) The thyroid gland increases secretion of T4 in order to cause a decrease in T3 secretion.

Answer C

Correct. Based on the figure, both thyroid hormones have an inhibitory effect on the hypothalamus; thus it would be predicted that less TRH would be secreted as T3 and T4 secretions increase.

85. Thyroxin is a hormone that increases metabolic activities within various tissue targets. Low levels of circulating thyroxin trigger the secretion of thyroid-stimulating hormone (TSH) from the anterior pituitary. TSH secretion then stimulates thyroxin production and release by the thyroid gland. The increased level of circulating thyroxin inhibits further secretion of TSH from the anterior pituitary.

Based on the information provided, which of the following can most likely be concluded about the TSH-thyroxin loop?

(A) A person taking thyroxin to supplement low thyroxin secretion will produce more TSH.
(B) Increased thyroxin production would cause elevated ribosomal activity in the anterior pituitary.
(C) The structure of the loop would lead to elevated thyroid and tissue activity due to positive feedback.
(D) The feedback mechanism would maintain relatively constant levels of thyroxin throughout tissue targets. ✓

4 Cell Communication and Cell Cycle MCQ

86. Researchers determined the average amount of time that a particular type of eukaryotic cell spends in each phase of the cell cycle. The data collected by the researchers are represented in Figure 1.

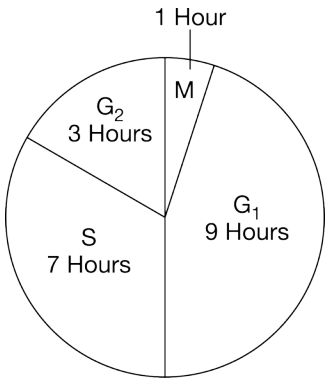


Figure 1. The average amount of time spent by a particular type of eukaryotic cell in each phase of the cell cycle

Based on Figure 1, what percent of the time required to complete a full cycle do the cells typically spend in interphase?

(A) 5%

(B) 35%

(C) 50%

(D) 95% ✓

Answer D

Correct. Interphase is composed of the G_1, S, and G_2 phases. Based on Figure 1, the cells typically spend 19 hours in interphase and require 20 hours total to complete one full cycle. Therefore, $\frac{19}{20} \times 100 = 95\%$.

87. Which of the following is true of mitosis?

(A) It is also known as cytokinesis.

(B) It maintains the same chromosome number in the daughter cells as in the parent cell. ✓

(C) It is the last phase of interphase.

(D) It regulates the transfer of genetic information from one daughter cell to another.

(E) It moves homologous chromosomes to opposite poles.

4 Cell Communication and Cell Cycle MCQ

88. Trypsinogen is split by the enzyme enterokinase to form an activated molecule of the protease trypsin. Which of the following would confirm that the activation of trypsin is an example of how a positive feedback mechanism can amplify a biological process?

(A) The activated trypsin enzyme can use enterokinase as a substrate.

(B) The trypsin produced by the reaction is capable of splitting and activating additional trypsinogen molecules. ✓

(C) If levels of trypsin were to get too high, the trypsin molecules would inhibit the enzyme enterokinase.

(D) Each mRNA molecule that codes for trypsinogen can be translated repeatedly to form many peptide molecules.

89. Type 1 diabetes results from the destruction of insulin-producing cells in the pancreas. Individuals with type 1 diabetes produce insufficient amounts of insulin, a hormone that regulates the concentration of glucose in the blood.

Which of the following best explains how treatment with a drug that stimulates the production of insulin receptors on target cells will affect the insulin signaling pathway in an individual with type 1 diabetes?

(A) The drug will have little or no effect on the signaling pathway because the receptors will not be activated in the absence of insulin. ✓

(B) The drug will have little or no effect on the signaling pathway because insulin receptors will not be allowed to enter the cells.

(C) The drug will restore the function of the signaling pathway because insulin levels will return to normal.

(D) The drug will restore the function of the signaling pathway because nonpancreatic cells will begin to produce insulin receptors.

90. Researchers studying cell cycle regulation in budding yeast have observed that a mutation in the *CDC15* gene causes cell cycle arrest in telophase when the yeast cells are incubated at an elevated temperature. Which of the following statements best predicts the effect of the cell cycle arrest on proliferating yeast cells?

(A) The yeast cells will transition out of G_0 but will fail to complete the G_1 phase.

(B) The yeast cells will initiate mitosis but will fail to complete the G_2 phase.

(C) The yeast cells will replicate their chromosomes but will fail to complete cytokinesis. ✓

(D) The yeast cells will replicate their organelles but will fail to complete the S phase.

Answer C

Correct. Because the cell cycle arrest occurs in telophase, the yeast cells will progress through the cell cycle, including replicating their chromosomes in the S phase, and will enter the M phase but will fail to complete mitosis and cytokinesis.

4 Cell Communication and Cell Cycle MCQ

91. A group of researchers cultured yeast cells in a nutrient-rich environment and a nutrient-poor environment and observed the duration of the stages of their cell cycles. The results of their study are summarized in Table 1.

Table 1. Duration (in minutes) of yeast cell cycle phases in a nutrient-rich environment and a nutrient-poor environment

Cell Cycle Stages	Nutrient-rich environment	Nutrient-poor environment
S and G_2	23	33
Mitosis	48	61
G_1	20	57
Cytokinesis	5	10

The cell cycle of yeast cells grown in the nutrient-poor environment is approximately what percent of the cell cycle of yeast cells grown in the nutrient-rich environment?

(A) 168 ✓

(B) 127

(C) 179

(D) 160

Answer A

Correct. Yeast cells in the nutrient-rich environment spend 96 minutes in the cell cycle compared to 161 minutes in the nutrient-poor environment. The percent can be determined by dividing 161 by 96, which results in 1.68 or 168 percent.

4 Cell Communication and Cell Cycle MCQ

92. A model of the typical life cycle of a cell is shown in Figure 1.

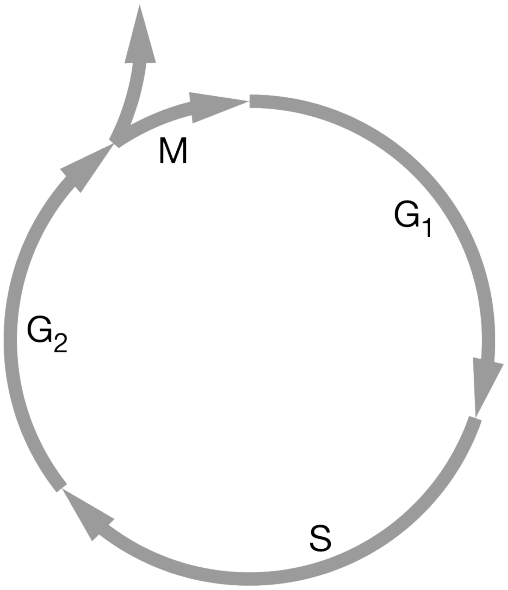


Figure 1. Typical life cycle of a eukaryotic cell

Scientists have estimated that it takes yeast cells approximately 20 hours to complete the entire cycle. Table 1 shows the amount of time in each phase of the life cycle for yeast cells.

Table 1. Amount of time spent in each stage of the cell cycle by yeast cells

Stage	M	G_1	S	G_2
Time (hours)	2	10	5	3

Based on Table 1, what percent of the life cycle of yeast cells is spent in DNA replication?

(A) 5 percent

(B) 10 percent

(C) 25 percent ✓

(D) 50 percent

Answer C

Correct. The yeast cells spend 5 hours out of a total of 20 hours in S phase, which is 25%.

4 Cell Communication and Cell Cycle MCQ

93. In mammals, an increase in the concentration of sodium in the blood triggers the release of antidiuretic hormone (ADH) from the pituitary gland. As the concentration of sodium in the blood returns to previous levels, the release of ADH from the pituitary gland is reduced. Based on the information presented, which of the following describes the most likely role of ADH in maintaining blood osmolarity?

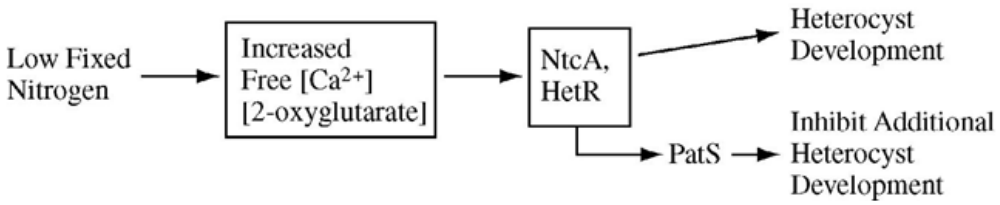
(A) ADH promotes an increase in the movement of sodium into the bloodstream.

(B) ADH promotes an increase in the movement of water into the bloodstream. ✓

(C) ADH promotes an increase in the excretion of water from the body.

(D) ADH promotes an increase in the secretion of additional ADH from the pituitary gland.

94. *Anabaena* is a simple multicellular photosynthetic cyanobacterium. In the absence of fixed nitrogen, certain newly developing cells along a filament express genes that code for nitrogen-fixing enzymes and become nonphotosynthetic heterocysts. The specialization is advantageous because some nitrogen-fixing enzymes function best in the absence of oxygen. Heterocysts do not carry out photosynthesis but instead provide adjacent cells with fixed nitrogen, in exchange receiving fixed carbon and reduced energy carriers.



As shown in the diagram above, when there is low fixed nitrogen in the environment, an increase in the concentration of free calcium ions and 2-oxyglutarate stimulates the expression of genes that produce two transcription factors (NtcA and HetR) that promote the expression of genes responsible for heterocyst development. HetR also causes production of a signal, PatS, that prevents adjacent cells from developing as heterocysts.

Based on your understanding of the ways in which signal transmission mediates cell function, which of the following predictions is most consistent with the information given above?

(A) In an environment with low fixed nitrogen, treating the *Anabaena* cells with a calcium-binding compound should prevent heterocyst differentiation. ✓

(B) A strain that overexpresses the *patS* gene should develop many more heterocysts in a low fixed nitrogen environment.

(C) In an environment with abundant fixed nitrogen, free calcium levels should be high in all cells so that no heterocysts develop.

(D) In environments with abundant fixed nitrogen, loss of the *hetR* gene should induce heterocyst development.

95. Cell communication is critical for the function of both unicellular and multicellular eukaryotes. Which of the following is likely true of cell signaling?

4 Cell Communication and Cell Cycle MCQ

(A) Cell signaling uses the highest molecular weight molecules found in living cells.

(B) Cell signaling has largely been replaced by other cell functions in higher mammals.

(C) Similar cell signaling pathways in diverse eukaryotes are evidence of conserved evolutionary processes. ✓

(D) Cell signaling functions mainly during early developmental stages.

96. An antigen can induce an immune response in a host organism. Antigens are targeted by antibodies that are produced by the organism's immune system in response to contact with the antigen. Antibodies are specific to antigens. Many different cell types are involved in immune responses.

Which of the following best explains the role of cell-to-cell communication during a response to an invasion of antigens?

(A) Chemicals that are secreted from antigen-presenting cells then activate helper T cells. ✓

(B) A macrophage cell engulfs a pathogen in the blood.

(C) Antigens attaching to receptors on memory B cells stimulate the memory B cells to become plasma cells.

(D) Antigen-presenting cells engulf antigens at the first exposure.

Answer A

Correct. Cell signaling pathways are activated by cell-to-cell communication.

97. Which of the following best explains how a group of cells can send chemical signals to cells in another part of the body?

(A) The chemical is released into the extracellular space and immediately binds to specific receptors on the surface of the target cells that released it.

(B) The chemical diffuses through the extracellular space until it reaches target cells with specific receptors for that chemical.

(C) The chemical is released into blood vessels, where it circulates until it reaches target cells that have specific receptors for that chemical. ✓

(D) The chemical is sent from the group of cells through neurons to the target cells, where it binds to specific receptors.

98. A cell signaling pathway begins when a hormone binds to its receptor. After binding, relay molecules within the cell become activated and trigger cellular responses. When drug K is added to cells, the cellular response is heightened and prolonged.

Which of the following mechanisms best explains how drug K increases the cellular response?

4 Cell Communication and Cell Cycle MCQ

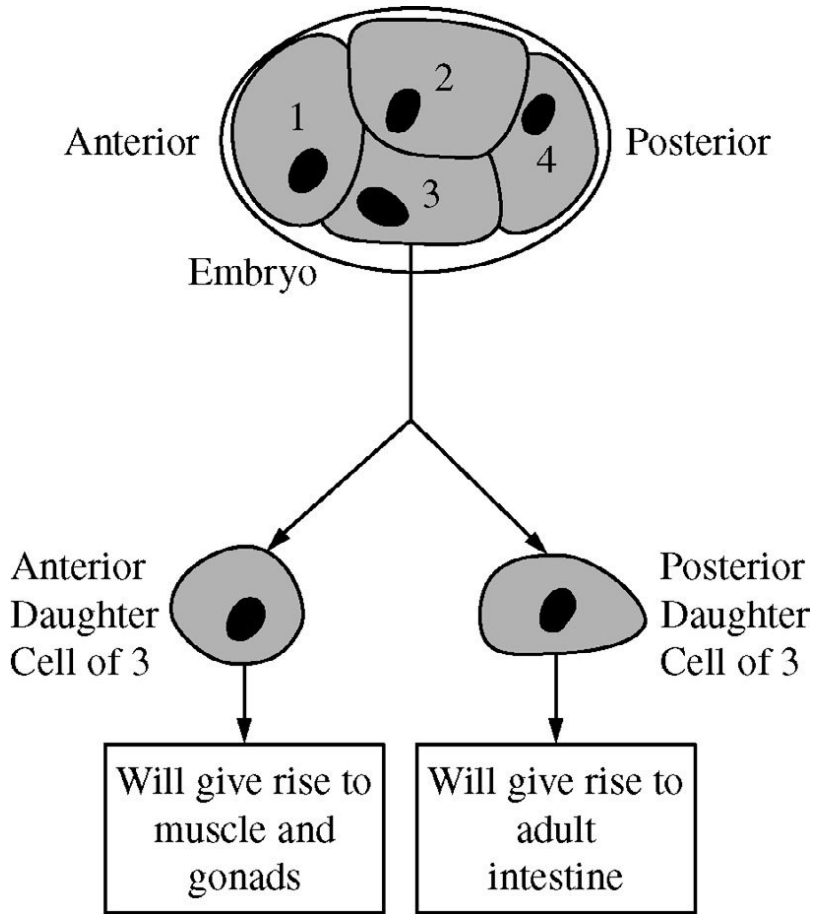
(A) Drug K inhibits the activation of the receptor protein.

(B) Drug K promotes the activation of relay molecules. ✓

(C) Drug K inhibits the deactivation of the receptor protein.

(D) Drug K promotes the circulation of relay molecules.

99.



The diagram above shows a developing worm embryo at the four-cell stage. Experiments have shown that when cell 3 divides, the anterior daughter cell gives rise to muscle and gonads and the posterior daughter cell gives rise to the intestine. However, if the cells of the embryo are separated from one another early during the four-cell stage, no intestine will form. Other experiments have shown that if cell 3 and cell 4 are recombined after the initial separation, the posterior daughter cell of cell 3 will once again give rise to normal intestine. Which of the following is the most plausible explanation for these findings?

(A) A cell surface protein on cell 4 signals cell 3 to induce formation of the worm's intestine. ✓

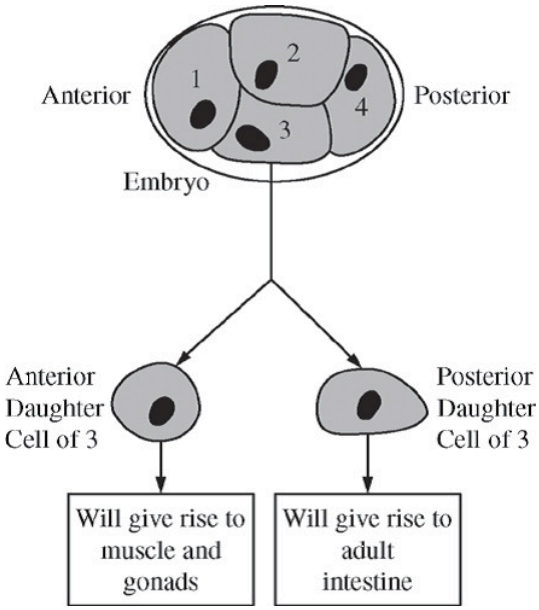
(B) The plasma membrane of cell 4 interacts with the plasma membrane of the posterior portion of cell 3, causing invaginations that become microvilli.

(C) Cell 3 passes an electrical signal to cell 4, which induces differentiation in cell 4.

(D) Cell 4 transfers genetic material to cell 3, which directs the development of intestinal cells.

4 Cell Communication and Cell Cycle MCQ

100.



The diagram above shows a developing worm embryo at the four-cell stage. Experiments have shown that when cell 3 divides, the anterior daughter cell gives rise to muscle and gonads and the posterior daughter cell gives rise to the intestine. However, if the cells of the embryo are separated from one another early during the four-cell stage, no intestine will form. Other experiments have shown that if cell 3 and cell 4 are recombined after the initial separation, the posterior daughter cell of cell 3 will once again give rise to normal intestine. Which of the following is the most plausible explanation for these findings?

(A) A cell surface protein on cell 4 signals cell 3 to induce formation of the worm's intestine. ✓

(B) The plasma membrane of cell 4 interacts with the plasma membrane of the posterior portion of cell 3, causing invaginations that become microvilli.

(C) Cell 3 passes an electrical signal to cell 4, which induces differentiation in cell 4.

(D) Cell 4 transfers genetic material to cell 3, which directs the development of intestinal cells.

Answer A

This option is correct. The stem of the question indicates that cell 4 and cell 3 must be touching in order for cell 3 to induce formation of the intestine. This is the mechanism by which the correct timing of this developmental event occurs.

101. A particular cell signaling pathway requires a hormone that is released into the bloodstream and travels to target cells. Scientists learned that not all cells respond to the hormone.

Which of the following best explains why some cells do not respond to the hormone?

4 Cell Communication and Cell Cycle MCQ

(A) Some cells lack hormone receptors. ✓

(B) Some cells lack regulators.

(C) Some cells lack cholesterol.

(D) Some cells lack embedded proteins.

102. Ethylene is an organic compound produced by ripening fruits. In a controlled experiment, researchers found that ethylene gas stimulated the ripening process in newly harvested fruits. Which of the following describes the most likely connection between natural ethylene production and fruit ripening?

(A) As a result of metabolic inactivity, newly harvested fruits are unable to absorb ethylene gas from the atmosphere.

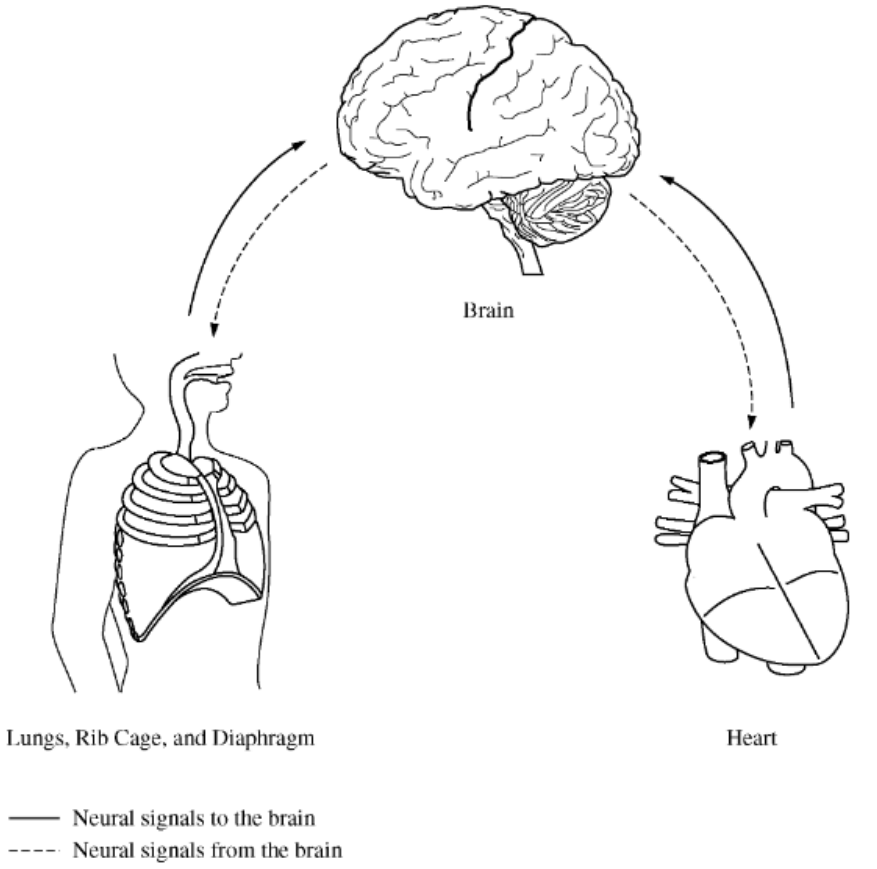
(B) Ethylene gas is a chemical signal through which ripening fruits trigger the ripening process in other fruits. ✓

(C) Because of normal phenotypic variation, only some of the fruits in a given generation are expected to produce ethylene gas.

(D) The rate of ethylene gas production by ripening fruits is an indicator of the relative age of an ecosystem.

4 Cell Communication and Cell Cycle MCQ

103. The brain coordinates the circulatory and respiratory systems of the human body. The control of breathing, for example, involves neural pathways among the structures represented in the figure above. One important stimulus in the control of breathing is an increase in blood CO_2 concentration, which is detected as a decrease in blood pH.



Which of the following best explains the physiological response to an overall increase in cellular respiration in the body?

(A) In response to depleted blood CO_2 levels, the pH sensors send signals directly to the rib muscles, resulting in an increase in the rate of CO_2 uptake by the lungs and a decrease in CO_2 utilization by the brain.

(B) In response to low blood pH, the pH sensors send a signal to the brain, which then sends a signal to the diaphragm, resulting in an increased rate of breathing to help eliminate excess blood CO_2 . ✓

(C) In response to high blood pH, the pH sensors send a signal directly to the lungs, resulting in a slower rate of breathing, and the lungs send a signal back to the heart once CO_2 availability has been restored.

(D) In response to an increased rate of breathing, the rib muscles send a signal to the brain, which then sends a signal to the heart, resulting in a decrease in heart activity and slower flow of blood through the body.

4 Cell Communication and Cell Cycle MCQ

104. Notch is a receptor protein displayed on the surface of certain cells in developing fruit fly embryos. Notch's ligand is a membrane-bound protein called Delta that is displayed on the surface of adjacent cells. When Notch is activated by its ligand, the intracellular tail of the Notch protein becomes separated from the rest of the protein. This allows the intracellular tail to move to the cell's nucleus and alter the expression of specific genes.

Which of the following statements best explains Delta's role in regulating cell communication through the Notch signaling pathway?

(A) Delta transmits a chemical signal to all the cells of a developing embryo.

(B) Delta allows the cells of a developing embryo to communicate without making direct contact.

(C) Delta restricts cell communication to short distances within a developing embryo. ✓

(D) Delta determines which cells in a developing embryo express the gene that encodes the Notch protein.

Answer C

Correct. Delta restricts cell communication to short distances by requiring direct cell-to-cell contact for activation of the Notch signaling pathway.

105. Vertebrate immune responses involve communication over short and long distances. Which of the following statements best helps explain how cell surface proteins, such as MHC proteins and T cell receptors, mediate cell communication over short distances?

(A) The proteins receive electrical signals from nerve cells.

(B) The proteins leave the cell and travel in the bloodstream to other cells.

(C) The proteins interact directly with proteins on the surfaces of other cells. ✓

(D) The proteins bind to molecules secreted by cells located in other parts of the body.

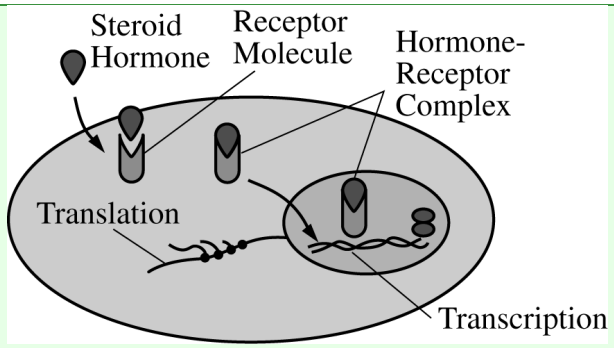
Answer C

Correct. In vertebrate immune responses, cell surface proteins, including MHC proteins and T cell receptors, mediate cell communication over short distances by interacting directly with proteins on the surfaces of other cells. In many cases, the protein-protein interaction activates signal transduction pathways that control specific cellular responses. The interaction by cell surface proteins is an example of cell communication that is limited to short distances because the communicating cells must be near to each other.

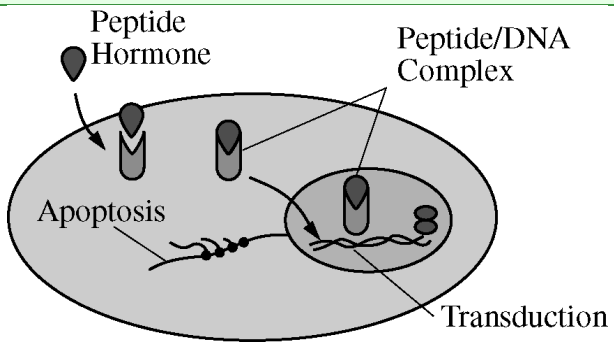
106. Which of the following diagrams best represents hormone-activated gene expression?

4 Cell Communication and Cell Cycle MCQ

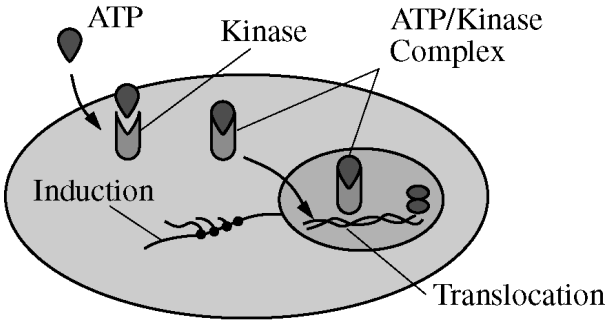
(A)



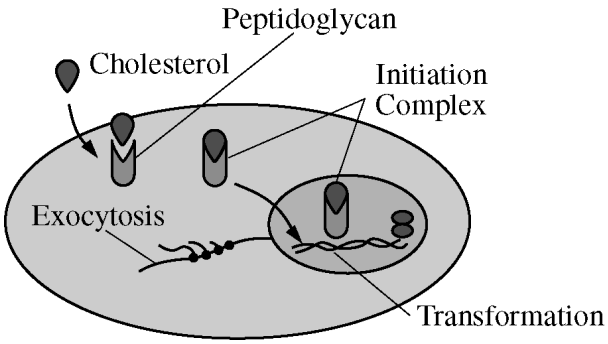
(B)



(C)



(D)



107. Which of the following best explains how small molecules move between adjacent cells in a plant shoot?

(A) The molecules are actively transported by motor proteins along the cytoskeleton.

(B) The molecules pass freely through plasmodesmata, which are cytoplasmic strands connecting two cells. ✓

(C) The molecules are swept along in the extracellular fluid by cilia projecting from cell membranes.

(D) The molecules bind reversibly to receptors on the cell membranes of xylem.

4 Cell Communication and Cell Cycle MCQ

108. In flowering plants, plasmodesmata are narrow channels through cell walls that connect the cytoplasms of adjacent cells. An explanation of how plant cells communicate across cell walls will most likely refer to the diffusion through plasmodesmata of which of the following?

(A) Membrane-bound organelles

(B) Condensed, duplicated chromosomes

(C) Branched polysaccharides

(D) Small, water-soluble molecules ✓

Answer D

Correct. Because plasmodesmata are narrow channels through plant cell walls, communication between adjacent cells most likely involves the diffusion of small, water-soluble molecules through the plasmodesmata, which connect the cytoplasms of the adjacent cells.

109. Adjacent plant cells have narrow channels called plasmodesmata that pass through the cell walls of the connected cells and allow a cytoplasmic connection between the cells.

Which of the following statements best explains a primary function of plasmodesmata?

(A) They allow the movement of molecules from one cell to another, enabling communication between cells. ✓

(B) They prevent the cell membrane from pulling away from the cell wall during periods of drought.

(C) They eliminate the need to produce signaling molecules and eliminate the need for cells to have receptors for signaling molecules.

(D) They increase the surface area available for attachment of ribosomes and thus increase protein synthesis.

Answer A

Correct. Plasmodesmata are channels through which cytosol exchange can occur, moving select molecules directly between cells. This enables cells enclosed by a cell wall to communicate.

4 Cell Communication and Cell Cycle MCQ

- **Directions:** Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

The following questions refer to the data in Figures 1 through 4 below, which were collected during a study of the growth of plant seedlings.

4 Cell Communication and Cell Cycle MCQ

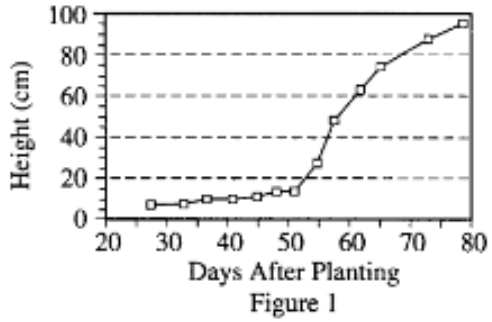


Figure 1

Figure 1 shows the average height of control seedings as a function of time after planting.

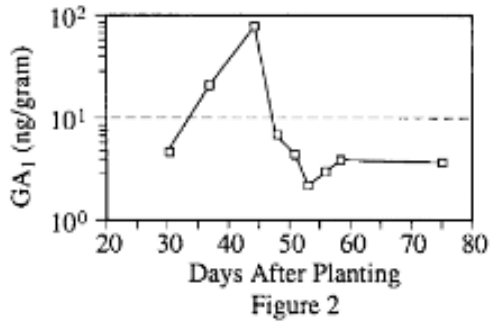


Figure 2

Figure 2 shows the concentration of one specific gibberellin compound, GA_1, that can be extracted from the plant seedling tissues (in units of nanograms of GA_1 per gram of plant tissue).

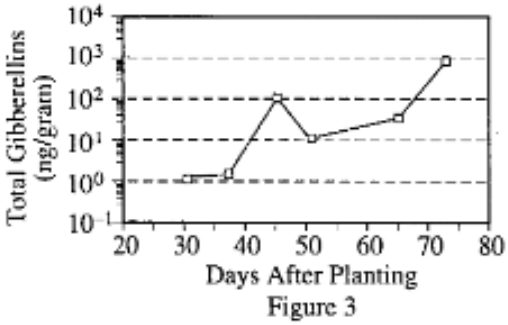


Figure 3

Figure 3 shows the concentration of all gibberellins that have physiological activity, including GA_1.

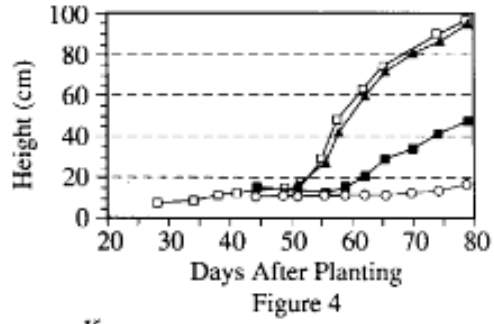


Figure 4

Figure 4 shows the patterns of growth that result when solutions containing different concentrations of an inhibitor of GA_1 synthesis are sprayed on the plant seedlings at regular intervals during the growth period.

Key

- ▫ Control seedlings not treated with inhibitor
- ▲ Seedlings treated with inhibitor at a concentration of 0.6 mg/ml
- ■ Seedlings treated with inhibitor at a concentration of 6.0 mg/ml
- ○ Seedlings treated with inhibitor at a concentration of 60.0 mg/ml

4 Cell Communication and Cell Cycle MCQ

110. When the concentration of GA_1 is highest, the average height of the control seedlings is approximately

(A) 10 cm ✓

(B) 30 cm

(C) 60 cm

(D) 85 cm

(E) 95 cm

111. The concentration of all gibberellins 65 days after planting is approximately

(A) 5 ng/gram

(B) 10 ng/gram

(C) 20 ng/gram ✓

(D) 120 ng/gram

(E) 150 ng/gram

112. Which of the following is a correct conclusion that can be drawn based only on the data in Figure 4?

(A) Seedling growth rates decrease between 50 and 80 days after planting at all concentrations of the inhibitor.

(B) The greater the inhibition of GA_1 synthesis, the lower the plant height after 80 days. ✓

(C) The spraying of GA_1 on seedlings results in an increase in seedling growth rates.

(D) The inhibitor kills the seedlings when it is applied in very high concentrations.

(E) The growth of the seedlings is directly proportional to the concentrations of inhibitor.

113. Which of the following is a correct conclusion that can be drawn from the data in Figures 1, 2, and 3?

(A) Most of the increase in the concentration of all gibberellins 40 to 45 days after planting is due to an increase in the concentration of GA_1. ✓

(B) The concentration of all gibberellins is three times as great at 75 days as it is at 30 days after planting.

(C) An increase in GA_1 levels occurs a day or two before the seedlings start to grow rapidly.

(D) An increase in the concentration of GA_1 inhibits the synthesis of other gibberellins.

(E) The concentration of GA_1 is a thousand times as great at 45 days as it is at 30 days after planting.

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114. G proteins are a family of receptor proteins that are involved in transmitting signals from outside a cell to inside a cell. When a signaling molecule binds to a G protein, the G protein is activated. The G protein then activates an enzyme that produces a second messenger called cAMP.

Which of the following describes a critical role of cAMP during the transduction stage of a G protein signal transduction pathway?

4 Cell Communication and Cell Cycle MCQ

(A) cAMP carries the signal to the nucleus of the cell and results in new sequences of nucleotides being added to the cell's DNA.

(B) cAMP binds the extracellular signal molecule and carries it to the intracellular target specified by the signal.

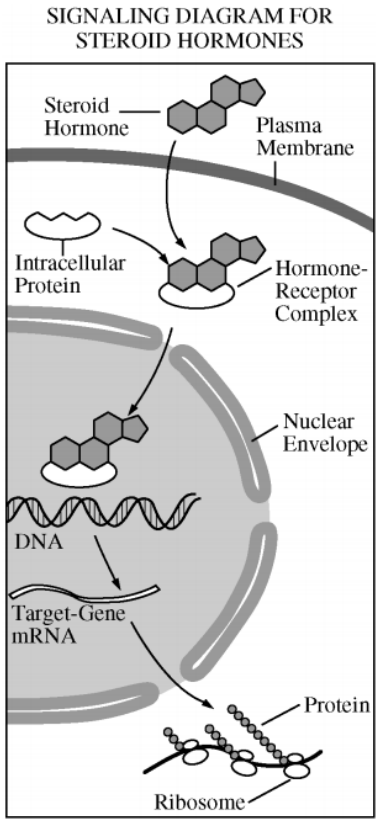
(C) cAMP modifies a specific monomer so that it can be added to an elongating structural macromolecule.

(D) cAMP results in the activation of an enzyme that amplifies the signal by acting on many substrate molecules. ✓

Answer D

Correct. Typically, cAMP immediately activates protein kinase A, which phosphorylates various other proteins. Kinases are a type of enzyme that activates proteins by transferring a phosphate group from ATP. Kinases are involved in many signaling pathways in a wide variety of organisms.

115. Steroid hormones, such as testosterone, pass through the plasma membrane and bind to an intracellular protein, as shown in the diagram below. The hormone-receptor complex then enters the nucleus, where it interacts with DNA to promote transcription of a specific gene.



Based on the information presented, which of the following will also occur in response to steroid signaling?

4 Cell Communication and Cell Cycle MCQ

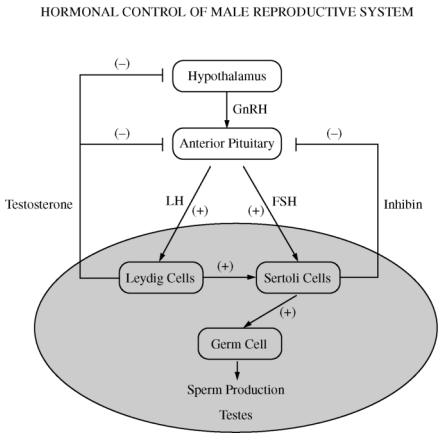
(A) Histone protein synthesis will increase because histones maintain the DNA in an optimal conformation for chromosome assembly.

(B) Ribosome production will increase because ribosomes are specific for the mRNA with which they bind during translation.

(C) DNA replication will increase as a result of the binding of the hormone-receptor complex to the DNA.

(D) Production of a specific mRNA will increase as a result of the binding of the hormone-receptor complex to the DNA. ✓

116.



Precise regulation of specific hormone levels is required for optimal sperm production in mammals, as summarized in the figure above. Anabolic-androgenic steroids (AAS) are synthetic variants of testosterone that are sometimes abused by persons who desire to enhance their athletic performance or alter their physique. Assuming that AAS function in the same way as naturally occurring testosterone, it is most likely that long-term abuse of AAS would

(A) stimulate FSH secretion

(B) stimulate testosterone production

(C) stimulate LH secretion

(D) reduce sperm production ✓

117. Insulin is a protein hormone that is secreted in response to elevated blood glucose levels. When insulin binds to its receptors on liver cells, the activated receptors stimulate phosphorylation cascades that cause the translocation of glucose transporters to the plasma membrane.

Based on the information provided, which of the following best describes the role of insulin in this liver cell signal transduction pathway?

(A) It acts as a ligand. ✓

(B) It acts as a receptor.

(C) It acts as a secondary messenger.

(D) It acts as a protein kinase.

118. Which of the following steps in a signaling pathway typically occurs first once a chemical messenger reaches a target cell?

4 Cell Communication and Cell Cycle MCQ

(A) Specific genes are activated.

(B) A second messenger molecule is produced.

(C) A ligand binds to a receptor. ✓

(D) Specific proteins are synthesized.

Answer C

Correct. The chemical messenger (ligand) first binds to a receptor, which then changes shape to initiate the signaling pathway.

119. A hydrophilic peptide hormone is produced in the anterior pituitary gland located at the base of the brain. The hormone targets specific cells in many parts of the body.

Which of the following best explains a possible mechanism that would enable the hormone to efficiently reach all of the target cells in the body?

(A) The hormone interacts with the nerves at the base of the brain and directs signals to the target cells through the nervous system.

(B) The hormone diffuses into target cells adjacent to the anterior pituitary gland, where the hormone is degraded.

(C) The hormone is released into the bloodstream where it can be transported to all cells with the correct receptors. ✓

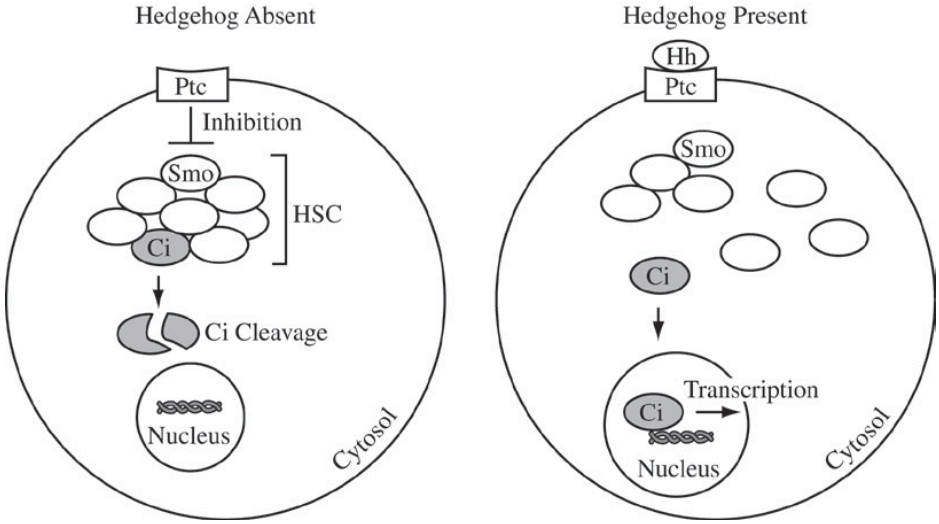
(D) The hormone moves through cytoplasmic connections between cells until it has reached all cells with the correct intracellular binding sites.

Answer C

Correct. The bloodstream is the most effective way to distribute a hormone that has to reach multiple target cells that are relatively long distances from the source of the hormone.

4 Cell Communication and Cell Cycle MCQ

120. The Hedgehog protein (Hh) plays a critical role during a certain period of embryo development, but it normally has no role in adults except for the maintenance of adult stem cells. However, the Hedgehog protein has been detected in 70 percent of pancreatic cancer cell samples. As illustrated in the figures below, the Hedgehog protein binds to an integral membrane protein receptor known as Patched (Ptc), thus initiating a pathway of gene expression. When Hedgehog is absent, Ptc inhibits another protein known as Smoothened (Smo), which, in turn, blocks the activation of a group of proteins collectively known as the Hedgehog signaling complex (HSC). The inactivation is the result of proteolytic cleavage of one component of the HSC complex, a transcription factor known as Cubitus interruptus (Ci). When Hedgehog is present, it binds to Ptc, which prevents the inhibition of Smo by Ptc. The result is that Ci remains intact and can enter the nucleus, where it binds to and activates certain genes.



One approach to treating patients with pancreatic cancer and other cancers in which the Hedgehog protein is detected is to modify the Hedgehog signaling pathway. Which of the following is the most useful approach?

(A) Treating patients with a molecule that is structurally similar to Hedgehog and that will bind to and interact with Ptc in the same fashion as Hedgehog

(B) Injecting patients with embryonic cells so that Hedgehog will bind to those cells instead of the cancer cells

(C) Treating patients with a membrane-soluble compound that can bind to Smo and block its activity ✓

(D) Injecting patients with a preparation of purified membrane-soluble Ci that will enter the nuclei of the cancer cells and induce gene transcription

121. Which of the following statements best describes how a growth factor stimulates cell division from outside a cell?

(A) The growth factor binds to other cells in the same area and holds them together to form a large, multicellular structure.

(B) The growth factor binds to receptors on the cell surface, initiating a signal transduction pathway that activates specific target genes. ✓

(C) The growth factor binds to sugar molecules in the extracellular fluid and provides them to the cell as a source of energy.

(D) The growth factor binds to phospholipids in the plasma membrane, creating a channel through which substances enter the cell.

4 Cell Communication and Cell Cycle MCQ

122. Over 100 different types of G-protein-linked receptors have been observed in eukaryotes, including scent receptors in animals, mating-signal receptors in yeasts, and light-sensing proteins in vertebrates. The proton pumps of bacteria have protein structures similar to G-protein-linked receptors. Based on this information, which of the following scientific questions will best guide research on the evolution of G-protein-linked signal transduction?

(A) Did the G-protein-linked signal transduction pathway in eukaryotes arise from a precursor in prokaryotes? ✓

(B) Are G-protein-linked receptors acquired from the environment as they are needed?

(C) Do eukaryotes require symbiotic prokaryotes in order to use G-protein-linked receptors?

(D) Do single-celled organisms have more efficient cell communication than do multicellular organisms?

Answer A

Correct. Prokaryotes have protein structures similar to eukaryotic G-protein-linked receptors. Since prokaryotes arose earlier in evolution, it is feasible to research the evolution of the G-protein-linked signal transduction pathway in eukaryotes as arising from a prokaryotic precursor.

123. Epinephrine is a protein hormone found in many animals. Epinephrine stimulates a signaling pathway that results in the breakdown of glycogen to glucose in the liver cells. Which of the following describes the initial steps in the process whereby epinephrine stimulates glycogen breakdown?

(A) Epinephrine binds to a cell-surface receptor; the activated receptor stimulates production of the second messenger, cAMP. ✓

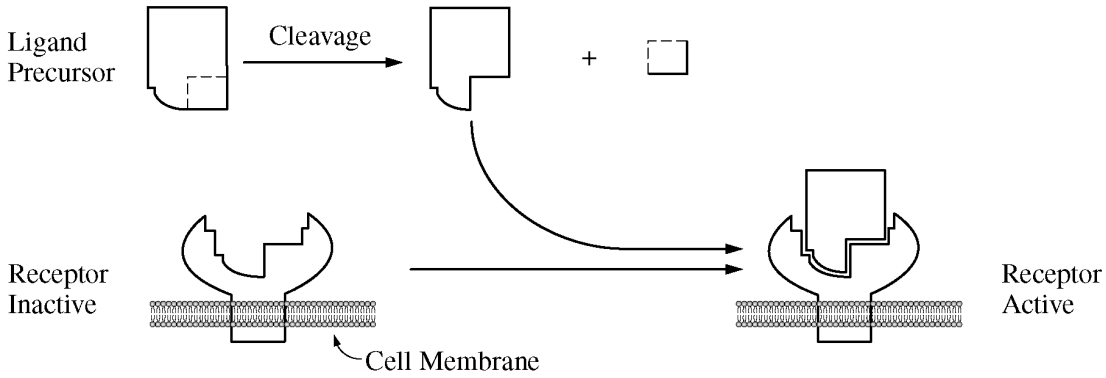
(B) Epinephrine binds to a cell-surface receptor; the activated receptor catalyzes the conversion of glycogen to glucose.

(C) Epinephrine diffuses through the plasma membrane; the hormone dimerizes in the cytosol.

(D) Epinephrine is taken into the cell by endocytosis; glycogen is converted to glucose in the endocytotic vesicle.

4 Cell Communication and Cell Cycle MCQ

124.



The figure above shows a model of a ligand precursor being cleaved to produce an active ligand that binds to a specific receptor. Which of the following is most likely to reduce the binding of the active ligand to its receptor?

(A) A change in the cytoskeletal attachment of transmembrane proteins

(B) The presence of a large amount of the precursor form of the ligand

(C) An increase in the ratio of the number of unsaturated to the number of saturated fatty acid tails of the membrane lipids

(D) A mutation in the receptor gene that causes a substitution of a charged amino acid for a nonpolar amino acid in the ligand binding site of the receptor ✓

125. Which of the following best describes a molecule that relays messages from receptors to other molecules only inside the cell?

(A) The molecule is a ligand.

(B) The molecule is a hormone.

(C) The molecule is a secondary messenger. ✓

(D) The molecule is a transport protein.

126. Metformin is a drug used to treat type 2 diabetes by decreasing glucose production in the liver. AMP-activated protein kinase (AMPK) is a major cellular regulator of glucose metabolism. Metformin activates AMPK in liver cells but cannot cross the plasma membrane. By blocking AMPK with an inhibitor, researchers found that AMPK activation is required for metformin to produce an inhibitory effect on glucose production by liver cells.

Which of the following best explains the component that metformin represents in a signal transduction pathway that regulates glucose production in the liver?

(A) It is a secondary messenger that amplifies a signal through a cascade reaction.

(B) It is a ligand that activates the signal transduction pathway of the activation of AMPK. ✓

(C) It is an allosteric regulator that binds to a crucial section of the DNA that makes the enzymes needed for glucose uptake.

(D) It is a competitive inhibitor that binds to glucose and prevents it from entering the cell.

4 Cell Communication and Cell Cycle MCQ

Answer B

Correct. Since metformin does not cross the membrane, it is likely a ligand that binds to a receptor on the surface of the plasma membrane, initiating the signal transduction pathway.

127. In a certain signal transduction pathway, the binding of an extracellular molecule to a cell-surface protein results in a rapid increase in the concentration of cyclic AMP inside the cell. The cyclic AMP binds to and activates cytosolic enzymes that then activate other enzymes in the cell.

Which of the following statements best describes the role of cyclic AMP in the signal transduction pathway?

(A) It acts as a signaling molecule that passes the signal from the cell to other cells.

(B) It acts as a receptor that carries the signal from outside the cell to inside the cell.

(C) It acts as a second messenger that helps relay and amplify the signal within the cell. ✓

(D) It acts as a channel protein that transmits the signal across the cell's nuclear membrane.

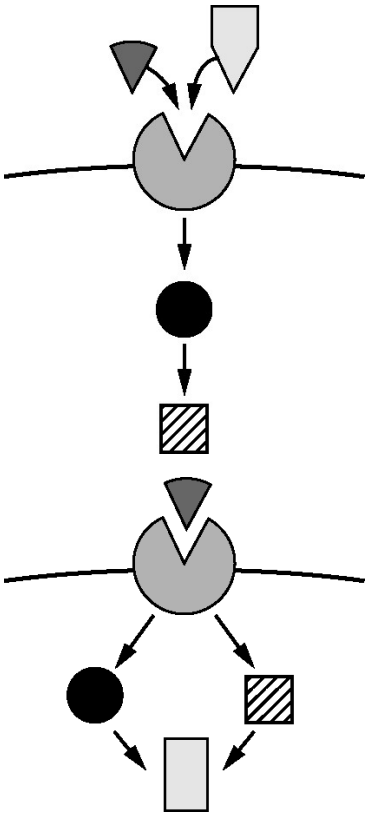
Answer C

Correct. In the signal transduction pathway, cyclic AMP acts as a second messenger by relaying the signal from the plasma membrane to cytosolic enzymes. The signal is amplified inside the cell by the production of many cyclic AMP molecules, which then activate enzyme molecules that are capable of catalyzing a specific reaction repeatedly.

128. Which of the following best represents two different signaling pathways that share a second messenger?

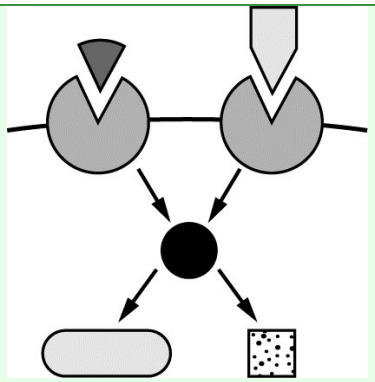
4 Cell Communication and Cell Cycle MCQ

(A)

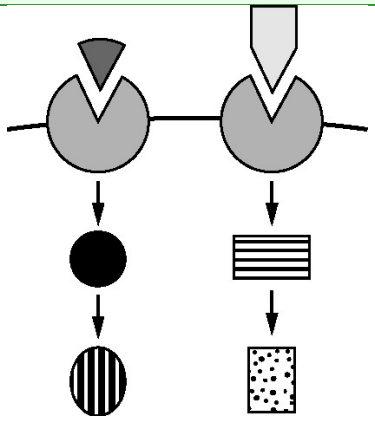


(B)

(C) ✓



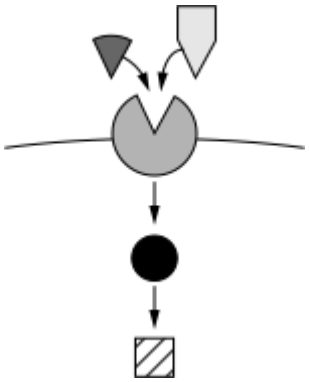
(D)



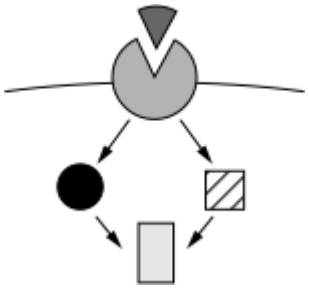
129. Which of the following best represents a signaling pathway that will lead to multiple responses from the same signaling molecule?

4 Cell Communication and Cell Cycle MCQ

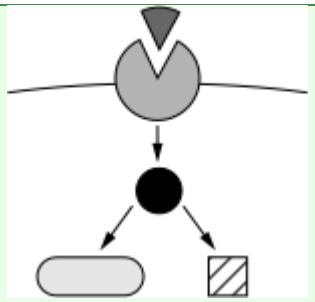
(A)



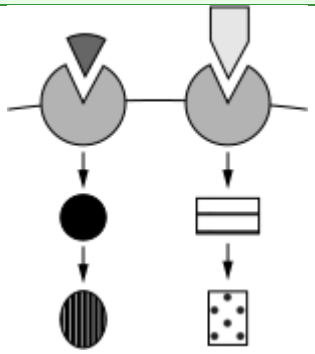
(B)



(C) ✓



(D)



Answer C

Correct. The initial signal activates an intermediary that activates two different pathways and produces two unique responses.